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Assessing the Progress of Stock Rebuilding in the Northeast Atlantic Against Levels That Can Produce Maximum Sustainable Yield

Henning Winker¹ | Massimiliano Cardinale¹  | Rishi Sharma²  | Laurence T. Kell³ | Iago Mosqueira⁴ | Christopher Griffiths¹ 

¹Department of Aquatic Resources, Institute of Marine Research, Swedish University of Agricultural Sciences, Lysekil, Sweden | ²Fishery and Aquaculture Division, Food and Agricultural Organization, Rome, Lazio, Italy | ³Centre for Environmental Policy, Imperial College London, London, UK | ⁴Wageningen Marine Research, IJmuiden, the Netherlands

Correspondence: Massimiliano Cardinale (massimiliano.cardinale@slu.se)

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ABSTRACT

Rebuilding fish stocks to levels above which they produce Maximum Sustainable Yield (MSY) is a management aim for all European commercially exploited stocks. Progress is typically monitored against the fishing mortality that produces MSY in the long term (F_{MSY}), however, the corresponding biomass target (B_{MSY}) is rarely evaluated nor reported. Here, we analyse a unique database of 73 quantitative ICES stock assessments to provide estimates of B_{MSY} across the Northeast Atlantic and apply a Bayesian state-space model to estimate joint trajectories of F/F_{MSY} and B/B_{MSY} . Our results confirm that median fishing mortality has substantially decreased from its peak in 1999 to just below F_{MSY} in 2020. Despite this, approximately half of the stocks remain fished above F_{MSY} , with 36% exceeding $1.2 \times F_{MSY}$. Biomass increased on average from below $0.5 B_{MSY}$ in 2000 to $0.68 B_{MSY}$ in 2020, but only 40% of stocks are currently above B_{MSY} and only 35% have an age structure that is comparable with fishing at F_{MSY} . Biomass relative to the ICES trigger point ($MSY B_{trigger}$) indicates that more than 70% of stocks are currently within safe biological limits. However, using $MSY B_{trigger}$ as a surrogate for B_{MSY} results in an over-optimistic classification of stock status, which conflicts with past levels of exploitation and may hinder stock rebuilding and the achievement of MSY objectives. Future projections from individual assessment forecasts predict further increases in B/B_{MSY} under current F levels. However, to achieve B_{MSY} by 2030, a 'perfect' implementation of the ICES Advice Rule would be required.

1 | Introduction

In 2002, the Johannesburg Declaration of the World Summit on Sustainable Development (WSSD; United Nations 2002) agreed that there was an urgent need to restore the world's depleted fish stocks to levels of biomass that can readily produce Maximum Sustainable Yield (MSY). Since then, MSY has become one of the main management targets when developing scientific advice frameworks. For example, in 2013, MSY was adopted within the European Common Fisheries Policy (CFP) (EU Regulation No 1380/2013; the Basic Regulation), with

Article 2(2) of the Basic Regulation stating that: 'the Common Fisheries Policy shall apply the precautionary approach to fisheries management and shall aim to ensure that exploitation of living marine biological resources restores and maintains populations of harvested species above levels which can produce the maximum sustainable yield. To progressively restore and maintain populations of fish stocks above biomass levels capable of producing maximum sustainable yield, the maximum sustainable yield exploitation rate shall be achieved by 2015 where possible and on a progressive, incremental basis at the latest by 2020 for all stocks'. This implies that there is

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an obligation for those acting under the Common Fisheries Policy framework (e.g., EU institutions and member states) to rebuild and maintain exploited fish stocks above biomass levels that produce MSY (Froese et al. 2018).

Reference points are central to fisheries management and are typically used to classify and communicate the current status of a fish stock. They are also used to determine the sustainable limits for future fishing opportunities such as the allowable catch level in quota-managed fisheries. Reference points are commonly defined as targets, thresholds and/or limits with respect to stock status. Target reference points represent a sustainable state of a fish stock that can produce a desirable long-term yield on average. By contrast, limit reference points are typically determined as a stock level below which productivity is impaired. Limit reference points therefore define the lower limit that should be avoided and not breached with high probability (Caddy and Mahon 2004). Threshold reference points typically serve as a precautionary buffer between the target and limit reference points. Threshold reference points can also coincide with trigger points that prompt an additional management response via a harvest control rule to help ensure that the fishery remains close to the target and avoids breaching the limit. The key difference is that target, threshold and limit reference points are benchmarks that determine the biological objectives for stock classification, whereas triggers are operationalised control points of a harvest control rule used to achieve the objectives (Cox et al. 2013).

Target reference points are often formulated as estimates or proxies of the fishing mortality (F_{MSY}) and biomass (B_{MSY}) that can produce MSY in the long term. Estimates of B_{MSY} are reported globally by many fisheries agencies and are frequently used for stock status classification (Hilborn et al. 2021; Sharma et al. 2020). For instance, the Food and Agriculture Organization of the United Nations (FAO) publishes a biannual report on the State of World Fisheries and Aquaculture (<https://www.fao.org/3/cc0461en/online/cc0461en.html>) and uses a biomass sustainability threshold of 80% of B_{MSY} to estimate the proportion of stocks that are considered overfished (FAO 2022; Sharma et al. 2025). This buffer is intended to account for natural fluctuations around B_{MSY} when fishing at or below F_{MSY} (Mace 2001). The Marine Steward Council (MSC) standard for certification and ecolabelling of sustainable seafood products has also adopted B_{MSY} as the default biomass reference point for scoring the stock status of a fishery resource (<https://www.msc.org/>).

B_{MSY} can be derived analytically as the biomass inflection point of the surplus-production curve where surplus production (i.e., yield) attains a maximum at equilibrium. The underlying assumption is that any changes in the biomass of a stock will emerge as a predictable response to regulation of fishing mortality, with the fishing mortality rate (F) that leads to B_{MSY} in the long term being defined as F_{MSY} . Derivation of F_{MSY} and B_{MSY} as part of a stock assessment are traditionally reported by tuna Regional Management Organizations (RMFOs), such as the International Commission for the Conservation of Atlantic Tunas (ICCAT) and the Indian Ocean Tuna Commission (IOTC). In Australia and the USA, biological proxies for F_{MSY} and B_{MSY} vary widely and include a fraction of the unfished biomass (B_0), or the

average biomass over years when the stock was considered close to B_{MSY} , but those based on per-recruit Spawning Potential Ratio (FSR_{PR}; typically FSR_{30%} to FSR_{45%}) and their corresponding biomass levels and/or B_{PR} (e.g., $B_{40%}$) are predominantly used for fisheries management and are considered to be more precautionary (Punt et al. 2014). In the New Zealand Harvest Strategy Standard (New Zealand Ministry of Fisheries 2008), detailed guidelines (New Zealand Ministry of Fisheries 2011) on selecting proxies, the so-called 'MSY compatible reference points', are specified for B_{MSY} as ratios to B_0 (i.e., B_{PR}), and for F_{MSY} based on FSR_{PR}. The default target is to achieve B_{MSY} (or its proxy) with at least 50% probability. Similarly, in Canada, F_{MSY} is determined analytically as the best estimate from the stock assessment model, and B_{MSY} is calculated as the expected long-term biomass corresponding to fishing at F_{MSY} (DFO 2009).

The International Council for the Exploration of the Sea (ICES; www.ices.dk), whose scientific advice framework is used to implement the EU's CFP, rarely reports implicit estimates of B_{MSY} (or a proxy) as B_{MSY} is not explicitly considered as a target reference point in the ICES MSY approach to fisheries management in the Northeast Atlantic. In fact, ICES does not formally consider any biomass target within its scientific advice framework, and instead a trigger and a limit reference point are used (ICES 2023a, 2023b). Within ICES, the Precautionary Approach (PA) to fishing (United Nations 1995; FAO 1995) was introduced in 1998 without initial consideration of explicit MSY based reference points, and the ICES MSY approach was only subsequently integrated into the PA framework in 2009 (ICES 2012). The ICES reference point system has evolved and undergone several revisions (Lassen et al. 2014; Silvar-Viladomiu et al. 2021). For quantitative analytical age-structured stock assessments in ICES (defined as Category 1 in ICES), F_{MSY} is commonly derived through stochastic forward simulations implemented in the Eqsim software (Simmonds et al. 2010). Within Eqsim, F_{MSY} is estimated under stochastic future conditions as the F that achieves maximum median long-term yield but is conditioned on the PA such that fishing at F_{MSY} cannot result in greater than the precautionary fishing mortality $F_{P0.5}$, which corresponds to 5% probability of the spawning stock biomass (SSB) falling below the limit reference point B_{lim} . B_{lim} in comparison is a deterministic biomass limit below which a stock is considered to have reduced reproductive capacity. For stocks where quantitative information is available, B_{lim} may also be identified as the stock size below which there is a high risk of reduced recruitment. Thus, the biomass that corresponds to fishing at F_{MSY} is an implicit output from Eqsim, but B_{MSY} is not reported within the current advice framework.

Stochastic forward projections are commonly run when assessing the appropriateness of the ICES Advice Rule (AR; ICES 2023a), by which fishing mortality is linearly reduced from F_{MSY} to zero if biomass falls below the ICES MSY $B_{trigger}$ point (hereafter $B_{trigger}$). In ICES, $B_{trigger}$ is formally defined as the lower bound of the SSB (Spawning Stock Biomass) fluctuation (5th percentile of the B_{MSY} estimate) when fishing at F_{MSY} . In practice, however, this $B_{trigger}$ definition is infrequently used (ICES 2022), and instead $B_{trigger}$ is usually set equal to the precautionary biomass reference point (B_{pa}), which is intended as a safety margin above B_{lim} that accounts for assessment uncertainty. For most stocks B_{pa} is approximated empirically by applying a multiplier to B_{lim} (typically $B_{trigger} \sim 1.4 B_{lim}$). ICES reports

B_{trigger} as the MSY biomass reference point for stock status classification and stocks with SSB larger than B_{trigger} are regarded by ICES as fluctuating around B_{MSY} . However, with B_{trigger} being defined as the lower bound of the SSB fluctuations around B_{MSY} , B_{trigger} is, by definition, not a target reference point but rather an operationalised management threshold that triggers a cautious reduction in fishing mortality. As such, B_{trigger} provides no means to assess whether stocks with a biomass larger than B_{trigger} are indeed fluctuating around B_{MSY} or are still rebuilding towards B_{MSY} with respect to global stock classification frameworks. An additional problem is that in the ICES framework, reference points are not estimated as part of the stock assessment process but by a post hoc procedure once the best assessment has been agreed. Fitting the stock recruitment relationship (SRR) externally using estimates of SSB and recruitment enables testing multiple candidate shapes and conducting sensitivity analyses without re-running the full assessment model. This two-stage procedure, however, does not automatically propagate uncertainty from the assessment into the SRR fit unless techniques like bootstrapping or Bayesian methods are employed, potentially underestimating risk and uncertainty. Inconsistencies can also arise when the assessment assumes a different (or no) SRR, creating risks of over-interpreting noise or artefacts due to autocorrelation or misspecification (ICES 2024a).

Although B_{trigger} is the lower bound of B_{MSY} and thus smaller than B_{MSY} , it is common practice to use B_{trigger} as a direct proxy for B_{MSY} for classification purposes when assessing stock status (e.g., Hilborn et al. 2020). For example, in the absence of B_{MSY} , the ICES B_{trigger} was also used by FAO's global fishery status classification for stocks in the Northeast Atlantic for its biennial report on the 'The State of World Fisheries and Aquaculture' (SOFIA; FAO 2022) and FAO's recent comprehensive 'Review of the state of world marine fisheries' (Sharma et al. 2025). On this, Hilborn et al. (2021) noted a strong anomaly in their results for groundfish stocks in the Northeast Atlantic. Specifically, the biomass started to exceed B_{MSY} in 2000 even though the average fishing pressure exceeded F_{MSY} , and this led the authors to conclude that either F_{MSY} is systematically underestimated or B_{trigger} may well be significantly below the biological B_{MSY} . In other studies, a discrepancy between B_{MSY} and B_{trigger} has also been noted, with approximations of B_{MSY} being either two times that of B_{trigger} (Froese et al. 2014) or three times that of B_{lim} (e.g., Sharma et al. 2020).

Twelve years after the adoption of MSY targets into the CFP, there is evidence that Europe has made substantial progress in reducing fishing pressure towards F_{MSY} for stocks in the Northeast Atlantic (e.g., Cardinale 2011, Cardinale et al. 2012; Fernandes and Cook 2013; Scientific, Technical and Economic Committee for Fisheries [STECF] 2023). There has also been a notably decreasing trend in fishing pressure in the Mediterranean Sea over the most recent decade, albeit a large proportion of stocks is still quite some distance from achieving sustainable exploitation levels in the region (e.g., Colloca et al. 2011; Cardinale et al. 2017; STECF 2023; FAO 2023). The Scientific, Technical and Economic Committee for Fisheries (STECF) is the scientific advisory body on fisheries policy to the European Commission and has the task of reporting on the CFP implementation through the estimation and publication of a series of indicators. However, progress with respect to rebuilding stocks to MSY levels is currently measured

against the input control F_{MSY} while the stock response in biomass relative to B_{MSY} is evaluated in the absence of formal B_{MSY} estimates for a large proportion of stocks. Therefore, the status of the Northeast Atlantic has never been formally assessed with respect to B_{MSY} , despite B_{MSY} being the overarching target of the CFP and FAO's SOFIA. This formal assessment of SSB against B_{MSY} is therefore a key research and management need in the evaluation of the CFP and the FAO's SOFIA, especially if reported progress is to be compared in a consistent and coherent way against policy objectives.

In this study, we used a unique database of 73 data-rich stocks assessed by ICES to evaluate the status of Northeast Atlantic stocks relative to the MSY targets of B_{MSY} and F_{MSY} . Given the importance of demography, that is, a stock's age and size structure, to stock rebuilding and productivity (e.g., Hixon et al. 2014; Barnett et al. 2017; Marshall et al. 2021), we also calculate and evaluate the age-based indicator AB_{MSY} , which informs on the age-structure of exploited fish stocks, and how the proportion of older fish corresponds to what is expected under F_{MSY} and at B_{MSY} (Griffiths et al. 2023). The addition of AB_{MSY} to evaluations of stock status based on B_{MSY} allows us to determine whether a stock has both the spawning biomass and the age structure it might need to sustain a certain level of catch (i.e., fishing at F_{MSY}). All results are presented in the form of the CFP, FAO and Kobe status classification frameworks meaning they are ready to be used in the evaluation of policy objectives. We also conduct future projections from individual assessment forecasts to the year 2030 and provide an outlook on the rebuilding of stock biomass to B_{MSY} levels under alternative scenarios with respect to the CFP. Throughout, model-based indicators are used to demonstrate the number of stocks that are above/below different reference points and are plotted through time to illustrate trends in exploitation, biomass and age structure. All indicators are modelled using a state-space model as implemented in the R-package JARA, which is used to monitor the performance of the CFP as required by Article 50 of the CFP (Gras et al. 2023).

2 | Materials and Methods

2.1 | Stock Assessment Database

A unique database of stock assessment inputs and outputs for 82 stocks covering the entire Northeast Atlantic was used in this study (Tables S1 and S2 modified from Griffiths et al. 2023 referred to hereafter as the ICES FLR stock assessment database). The database is composed of objects of the 'FLStock' class as defined in the FLR framework (Kell et al. 2007; ICES 2022) and has recently been used by Griffiths et al. (2023), Moesgaard Albertsen et al. (2024) and Cardinale et al. (2024). All 82 stocks are classified and assessed by ICES as Category 1 stocks (i.e., age-based analytical stock assessments) and cover an assessment horizon with final data years until 2019 ($n=12$), 2020 ($n=63$), 2021 ($n=6$) and 2022 ($n=1$). In the following text, stocks are referred to by their ICES stock IDs (<https://sid.ices.dk>), with details on the assessment inputs and outputs of all stocks provided in the form of an external open access Shiny application (<https://maxcardinale.shinyapps.io/Indicators/>). The Shiny App also contains a range of plots that visualise various aspects of each stock's population dynamics and demographic characteristics.

The database contains 25 bony fish species (representative of 12 taxonomic families and nine taxonomic orders) as well as one crustacean, *Pandalus borealis* (pra.27.3a4a) (Figure 1). It also contains harmonised assessment inputs and outputs from 12 different age-structure stock assessment frameworks of which SAM ($n = 35$; Nielsen and Berg 2014) and Stock Synthesis (SS3; $n = 15$; Methot and Wetzel 2013) are the most frequent (Table S3 modified from Griffiths et al. 2023). The majority of the stocks occur in the North Sea ($n = 20$) and surrounding areas (e.g., Celtic Sea [$n = 8$] and Irish Sea [$n = 7$]); however, some stocks have a wide spatial distribution that spans multiple ICES ecoregions (e.g., Blue whiting, *Micromesistius poutassou*, in the Northeast Atlantic and adjacent waters; whb.27.1-91214).

For each stock, reference points were updated using the official estimates for F_{MSY} , $B_{trigger}$ and B_{lim} from the most recent ICES advice (Table S2). In the FLStock objects, mean fishing mortality (F_{bar}) is computed as average F -at-age across a range of representative age-classes and was set during each stock's benchmark assessment. For stocks for which fishing pressure is presented as harvest rate (i.e., cod.27.5a, had.27.5a and pok.27.5a), the harvest rate (HR) was translated into F . This was done by first calculating the annual average F -at-age (F_{bar}) and plotting it

against yearly HR. A linear relationship was then fitted between F_{bar} and HR to estimate F_{bar} corresponding to the stock-specific HR reference point of ICES.

2.2 | Stochastic Equilibrium Estimation of B_{MSY}

As not all stocks have formally accepted F_{MSY} reference points (Table S2), B_{MSY} could only be derived for 73 stocks. Thus, nine stocks were removed from our analysis. This means that the final database contains 73 stocks belonging to 24 species, which occur in 13 different areas and are assessed using 11 different assessment models. The database covers most of the species and stocks that are assessed by ICES as Category 1 and includes a range of analytical models as well as most geographical areas.

Silvar-Viladomiu et al. (2022) characterised the available documentation on the benchmark settings used for reference point estimation as difficult to retrieve, insufficiently documented, with a specification of the SRR type missing for more than 50% of the stocks. In this context, it is therefore important to stress that the aim here is not to produce an exact estimate of B_{MSY} , which would depend heavily on stock-specific assumptions

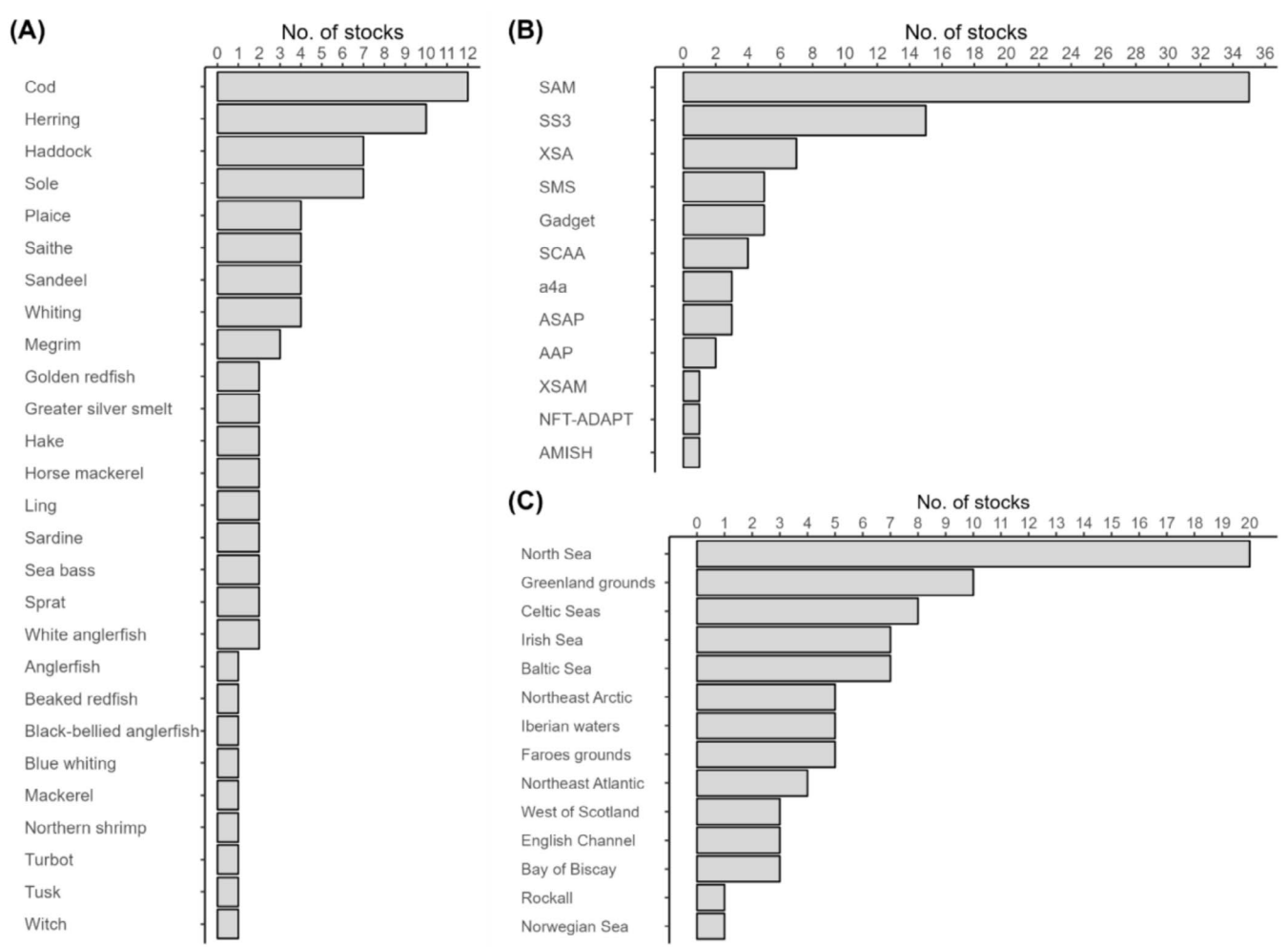


FIGURE 1 | Graphical summary of the original database used in this study. The database contains 26 species (A) from 14 areas (B), whereas the area is a broad characterisation based on ICES areas and ecoregions. All 82 stocks are assessed using age structured analytical assessment frameworks (C), the details of which can be found in Tables S1 and S3. Each stock's SSB and F values in the final year of the assessment are presented in Table S1, and stock-specific reference points are detailed in Table S2.

agreed at stock-specific benchmarks, but instead to provide a general, transparent and reproducible comparison of MSY , $B_{trigger}$ and B_{MSY} to investigate what effect it has on stock status classification across a range of stocks.

To obtain consistent estimates of B_{MSY} , we ran the ICES stochastic equilibrium reference point software Eqsim (e.g., ICES 2017; <https://github.com/wgmg/msy>) with standard sets of the commonly used default configurations for all 73 stocks. Eqsim provides MSY reference points based on the equilibrium distribution of stochastic projections, and the B_{MSY} estimate of interest can be routinely computed as the SSB estimate corresponding to fishing at a given F_{MSY} value.

For each stock, simulations were run for 200 years with the last 50 years retained to compute equilibrium values from the simulated distributions, which is ICES standard procedure when estimating reference points with Eqsim. The stochastic long-term projections require users to make a choice about the functional form (i.e., segmented regression, Beverton and Holt, Ricker) of the SRR. Although structural uncertainty about the SSR can also be considered by applying model averaging using smooth AIC weights (Buckland et al. 1997), the use of single segmented regression SRRs tends to be the predominant choice. For example, the 2015 ICES workshop on MSY based reference points (ICES 2017) almost exclusively used segmented regression SSRs, with very few exceptions. Accordingly, we used a segmented regression SRR as the standard assumption for all stocks:

$$R_{t+lag} = \begin{cases} aSSB_t & \text{if } SSB_t < b \\ ab & \text{if } SSB_t \geq b \end{cases} \quad (1)$$

where SSB_t is the spawning biomass at year t , lag denotes the age of new recruits R , a is slope between zero and the break-point b , which was fixed for each stock to $b = B_{lim}$ consistent with the 2023 ICES advice (ICES 2023a). The asymptotic recruitment R_0 is given by the product of ab .

In addition to this, we also evaluated the sensitivity of B_{MSY} estimates to assuming a functional form of the Beverton and Holt stock recruitment relationship:

$$R_{t+lag} = \frac{aSB_t}{b + SBB_t} \quad (2)$$

Initial trials indicated that the scale was in many instances poorly informed for many stocks due to lack of contrast in the available SSB time series, which can be addressed by penalising R_0 as a parameter to avoid spurious and likely implausible stock recruitment relationships (Moesgaard Albertsen et al. 2024). For the purpose of the sensitivity, we therefore used the FLR library FLSRTMB (<https://github.com/flr/FLSRTMB>) to fit a re-parameterised Beverton and Holt SRR as a function of steepness s and annual unfished spawning biomass per-recruit SPR_0 (Mace and Doonan 1988):

$$R_{t+lag} = \frac{4sSB_tR_0}{R_0SPR_0(1-s) + SB_t(5s-1)} \quad (3)$$

where steepness s is defined as the ratio of recruitment, when SSB equals 20% of B_0 , to the unfished recruitment R_0 , and SPR_0 is the unfished spawning biomass per recruit computed as a function of the average annual quantities of weight at age, maturity at age and natural mortality at age. The maximum likelihood estimate of R_0 was penalised by invoking a lognormal prior with a CV of 30% and the mean corresponding to the natural logarithm of the R_0 estimate from the segmented regression (Equation 1).

We then compared the equilibrium estimates of B_{MSY} obtained using the segmented regression SRR ($B_{MSY} SR$), B_{MSY} obtained using a Beverton and Holt SRR ($B_{MSY} BH$) (both using ICES official F_{MSY} estimates) and B_{MSY} obtained using a Beverton and Holt SRR and using the F_{MSY} estimated directly to maximise the production curve ($B_{MSY} Est$). The comparison showed a high level of correlation between the different estimates (Figure S1; Table S2), and therefore B_{MSY} estimated from a segmented regression using ICES official F_{MSY} estimates (i.e., $B_{MSY} SR$) is used throughout our analysis.

Stochastic recruitment is simulated by resampling from the predictive distribution of the segmented regression or Beverton and Holt fit to each stock's SRR. For the assessment error in the advisory year and the autocorrelation in the assessment error in the advisory year, ICES default values (i.e., 0.212 and 0.423, respectively) were used (ICES 2015). In all Eqsim simulations, recruitment variation and a first-order autocorrelation (AR1) coefficient were estimated from observed recruitment residuals and propagated into the future recruitment simulations. All stock parameters (i.e., weight-at-age, maturity-at-age, natural mortality at age and selectivity-at-age) were set as the average of the last 3 years for all stocks. The reason for the choice was to ensure that the B_{MSY} estimates reflect recent biological and selectivity conditions to generate management advice that is robust to potential non-stationarity in stock dynamics, consistent with the 'dynamic B_0 ' concept (Bessell-Brown et al. 2022). Analyses from the ICES workshop WKREF1 (ICES 2022) demonstrated that forecasting bias increases when reference points are estimated over longer historic windows, as they may not capture recent changes in population productivity or fishery selectivity. Forecast bias was minimised when the estimation window was restricted to the most recent 3 years (ICES 2022). This choice therefore supports the aim of providing advice that is both responsive to current conditions and is as predictive as possible when it comes to future stock trajectories, in line with evolving ICES best practice for dynamic reference points (e.g., ICES 2023b). An additional reason for this choice relates to the relevance and generality of the analysis through time. The aim is not to recreate the exact conditions used in each benchmark, which is difficult, as the estimates would only be relevant by stock and until the next benchmark. Instead, using a standard procedure (3-year average for biology parameters and selectivity, and a standard SRR) we are able to make comparisons across all stocks. The quantity of interest, B_{MSY} , was then derived as the long-term median of the SSB distribution derived from stochastic projections that were run with a fixed F set to the official F_{MSY} estimate for each stock.

2.3 | Stock Status Indicators

To evaluate the status of fish stocks in the Northeast Atlantic, we selected the following four stock status indicators: (1) F/F_{MSY} , (2) B/B_{MSY} , (3) B/MSY $B_{trigger}$ and (4) ABI_{MSY} . These indicators relate to fishing mortality, biomass and age structure, respectively, and are all reported on a relative scale (to their specific reference points), allowing them to be combined and compared across stocks.

Age or size structure is not currently considered in the evaluation of the CFP or FAO's SOFIA; however, it has been identified as an important component of the Marine Strategy Framework Directive's (MSFD) Descriptor 3 (D3) of Good Environment Status (GES), which is considered an overarching policy for exploited fish stocks in the European Union. The MSFD D3's stipulates that the status of exploited fish stocks should be evaluated based on three criteria: (1) fishing mortality, (2) spawning stock biomass and (3) age or size structure, and that GES should have been achieved by 2020 (EU-COM 2008, 2017). Thus, we also consider it here and have chosen to use ABI_{MSY} over other age or length-based indicators (e.g., 95% percentile of the fish length distribution, L_{95} ; Probst et al. 2021) as tests show that it is both responsive to overfishing and generally tracks changes in B/B_{MSY} with high classification skill (Griffiths et al. 2023). Moreover, ABI_{MSY} is uniquely placed as it has a reference point and target that are linked to MSY and therefore integrates well with evaluations of stock status based on F/F_{MSY} and B/B_{MSY} (ICES 2024b).

ABI_{MSY} was estimated using the methods detailed in Griffiths et al. (2023). In short, this involves running stock-specific long-term deterministic forecasts (for ~200 years) with a constant F that is equal to F_{MSY} (detailed in Table S2). Once equilibrium is attained in terms of biomass, we then extract the corresponding age structure (numbers-at-age distribution) at B_{MSY} and calculate A_{MSY} and P_{MSY} . These variables correspond to the integer age that is nearest to the 90th percentile of the numbers-at-age distribution (A_{MSY}) and the realised percentage above A_{MSY} (P_{MSY}). This means that P_{MSY} won't always be exactly 10% but we expect it to be highly comparable. Prior to calculating A_{MSY} and P_{MSY} , we also removed the first age class from the numbers-at-age distribution to limit the potent impact of incoming recruitment on ABI_{MSY} .

In any given year t , we then identify A_{MSY} in the estimated number-at-age distribution and calculate the percentage of fish above it, P_t . ABI_{MSY} is then calculated as a relative indicator, such that:

$$ABI_{MSY,t} = P_t / P_{MSY} \quad (4)$$

and informs on how the current or past age structure of a stock in terms of older and larger fish compares to what should be expected when fishing at F_{MSY} in the long term.

As in the stochastic Eqsim forecasts used to estimate stock-specific B_{MSY} values, we make comparable assumptions of each stock's biology and fishing selectivity when estimating a stock's age structure at equilibrium. We set weight-at-age, maturity-at-age, mortality-at-age and selectivity to the average of the last 3 years.

2.4 | Estimation of Stock Status Indicators

To estimate joint trends (across stocks) in the stock indicators for biomass, fishing mortality and age structure, we used the R package *JARA* (Winker et al. 2020) to conduct a Bayesian State-Space trend analysis, which is available on GitHub (<https://github.com/henning-winker/jara>).

The joint trends for all four stock status indicators (F/F_{MSY} , B/B_{MSY} , B/MSY $B_{trigger}$ and ABI_{MSY}) were estimated for the period 1970–2020, as few stocks had data predating 1970, and data are only available up to 2020 for most stocks in our database (77%). Moreover, the CFP and MSFD both had stock status evaluation deadlines for 2020; therefore, providing a suitable cut-off for our analysis.

JARA was originally developed to provide a generalised Bayesian state-space decision-support tool for trend analysis of abundance indices with direct applications to IUCN Red List assessments (Sherley, Crawford, et al. 2019; Sherley, Winker, et al. 2019; Winker et al. 2020; Pacoureau et al. 2021). The implemented linear state-space modelling framework has been widely used to analyse the trends as evolving latent states in time series of animal population (e.g., Buckland et al. 2004). Specifically, we applied a recent extension of *JARA* that has been developed for CFP monitoring of fishing indicators and enables simultaneous fitting of multiple stock time-series with stock-specific trends and integration into a joint posterior (Winker et al. 2022; Gras et al. 2023). For this analysis, the main advantage of the state-space approach over simple time-series averaging is that the underlying state-process can make stock-specific predictions (forward and backward) for missing data, and it is based on inference from the stock-specific trend estimates while maintaining the ability to propagate the uncertainty in the combined estimates (Winker et al. 2022). This way, the state-space approach can correct for years with incomplete data coverage (Hilborn et al. 2020; Winker et al. 2022). Details on the *JARA* formulation are provided in the [Supporting Information](#).

For the Bayesian implementation, Monte-Carlo Markov Chain (MCMC) simulations were run using three MCMC chains, each with 9000 iterations, of which the first 2000 were discarded as 'burn-in' and thereafter every second iteration was saved (thinning). This yielded a total of 10,500 MCMC iterations to represent the posteriors' distributions.

The MCMC samples account for uncertainty about states in the time series for each stock. However, by computing the mean across the time series of stock trajectories this does not account for the property of the available stock assessments being a sample of a 'population' of stocks, which may change from year to year due to assessments being not updated or new assessments becoming available. To account for this, a non-parametric bootstrap procedure of resampling from the posteriors of stock-specific trajectories was applied after the MCMC to provide estimates of the global geometric mean and the associated uncertainty of the joint index across stocks (Winker et al. 2022). This assumes that the available sets of stock-specific time series represent a random sample of a population of stocks (see [Supporting Information](#)). Considering that the analysis covered the period 1970–2020, but some assessment time-series started only in the year 2000, trend-based back-casting over up to three decades was deemed to not be sufficiently

robust for these stocks. We therefore restricted the resampling of posteriors prior to 2000 to years for which the posteriors of a stock trajectory were informed by an observation. Note that for years with complete data coverage across stocks, the median of stock status trajectories is an unbiased estimator of the geometric mean of all observations for this year. In addition, the proportions that are above a given reference point (e.g., B_{MSY} or F_{MSY}) are also estimated from the non-parametric bootstrap procedure.

2.5 | Stock Status Classifications

To place our indicator estimates in the context of current policy objectives, we have adopted the status classification frameworks commonly used in the evaluation of the CFP and FAO's SOFIA. Specifically, for the CFP we present trends in *JARA* outputs (for all four stock status indicators) from 1970 to 2020 and demonstrate how both the estimated median ratio and the proportion of stocks above or below the respective reference points (F_{MSY} , B_{MSY} and $\text{MSY } B_{\text{trigger}}$) have changed through time. For the FAO, we also evaluate stock status (in terms of spawning stock biomass) based on the biomass sustainability threshold of 80% of B_{MSY} which is used to determine whether a stock is considered overfished or not (FAO 2022; Sharma et al. 2025).

We have also chosen to present taxonomic and region-based estimates, thus testing if stock status classification differs based on space and phylogeny. To do so, we run *JARA* models for specific taxonomic orders and geographical regions, specifically the three most common taxonomic orders (*Gadiformes*, *Pleuronectiformes* and *Clupeiformes*) and the four main geographical regions (Greater North Sea, Baltic Sea, Arctic waters and Celtic Seas) in our database. For further details on taxonomic orders and geographic regions, see Table S1.

We also present our results in the form of Kobe plots (Kell et al. 2016; also called the phase plot), which provide an intuitive way of visualising stock status in the context of biomass and fishing mortality. The Kobe plot is based on the concept that if fishing mortality (F) is above F_{MSY} , overfishing is judged to be occurring, whereas if the biomass (B ; or some measure of spawning output) is below B_{MSY} , the stock is judged to be overfished. Specifically, the Kobe plot plots B/B_{MSY} on the x-axis and F/F_{MSY} on the y-axis such that vertical and horizontal lines at 1.0 split the plot into four sections. The upper left represents a phase which is not desirable: overfishing occurring and an overfished stock; and the lower right represents a healthy stock: overfishing not occurring and an underfished stock. The trajectory of the stock over time is plotted such that the historic status of the stock can be tracked up to the most recent assessment. Typically, a stock starts in the lower right as the fishery develops, then moves anti-clockwise into the upper left as the population becomes overexploited, and finally, as appropriate management is applied, it continues to cycle around the centre of the plot (i.e., a point corresponding to B_{MSY} , F_{MSY}).

2.6 | Projections to 2030

To investigate how stock status may evolve in the short to medium term, and to provide an outlook of stock rebuilding in the Northeast Atlantic, we used a two-stage approach by first

forecasting the individual age-structured stock dynamics until 2030, using the 'FLasher' package within FLR, and then fitting *JARA* to the resulting four trajectories of all four stock status indicators (F/F_{MSY} , B/B_{MSY} , $B/\text{MSY } B_{\text{trigger}}$ and ABI_{MSY}). All stock-specific forecasts were conducted with a segmented regression SRR that was parameterised to be consistent with the Eqsim forecasts, with a fixed breakpoint set at B_{lim} while also setting weight-at-age, maturity-at-age, mortality-at-age and selectivity to the average of the last 3 years.

We consider the following four alternative forecast scenarios:

1. F_{sq} : flat forecast with F fixed at the current F for each stock.
2. AR 2023: A closed-loop forecast with a feedback control of the ICES Advice Rule from 2023, where F_{bar} in the intermediate years of 2021 and 2022 is set at F_{sq} .
3. AR 2014: A backtest approach (Kell et al. 2021) of applying feedback control loop with the ICES Advice Rule from 2014 onward, using hindcasts with the expected recruitment computed as a function of the SRR and estimated recruitment deviations through to 2020 and deterministic forecasts after that.
4. AR B_{MSY} : Identical to AR 2023, however, $\text{MSY } B_{\text{trigger}}$ is replaced by B_{MSY} in the ICES Advice Rule from 2023.

In each case, F_{sq} (status quo fishing mortality) refers to fishing at the F estimated for each stock in the most recent assessment year. In comparison, all three AR scenarios assume a perfect implementation of their respective ICES Advice Rules with feedback control, where F is at F_{MSY} only if a stock's biomass exceeds B_{trigger} or B_{MSY} and is linearly decreased towards zero otherwise. In each case, the estimated ratio for F/F_{MSY} , B/B_{MSY} , $B/\text{MSY } B_{\text{trigger}}$ and ABI_{MSY} with uncertainty was calculated and extracted for all 73 stocks. These are then combined into a joined posterior of a 'global' stock status trajectory using *JARA* as described above and presented as both the estimated median ratio and the proportion of stocks above or below the respective reference points with the associated uncertainty across stocks.

As is the case for our stock status evaluations for the period 1970–2020, we also evaluate each forecast scenario through 2030 using the stock status classifications of the CFP and the FAO. We have also chosen to evaluate forecasts of ABI_{MSY} against the threshold of 0.8, as it matches the B/B_{MSY} sustainability threshold used by the FAO's SOFIA and is used in simulation testing detailed in Griffiths et al. (2023).

3 | Results

3.1 | Trends in Fishing Mortality, Biomass and Age Structure

The estimated combined median ratio of F/F_{MSY} , B/B_{trigger} , B/B_{MSY} and ABI_{MSY} is shown in Figure 2, and on a stock-by-stock basis in Figure S2. The estimated ratios indicate an initial decline in SSB coinciding with an increase in F . As fishing pressure started to decline from $F/F_{\text{MSY}} > 1.5$ to a value of 0.91 in 2020, SSB started to recover continuously.

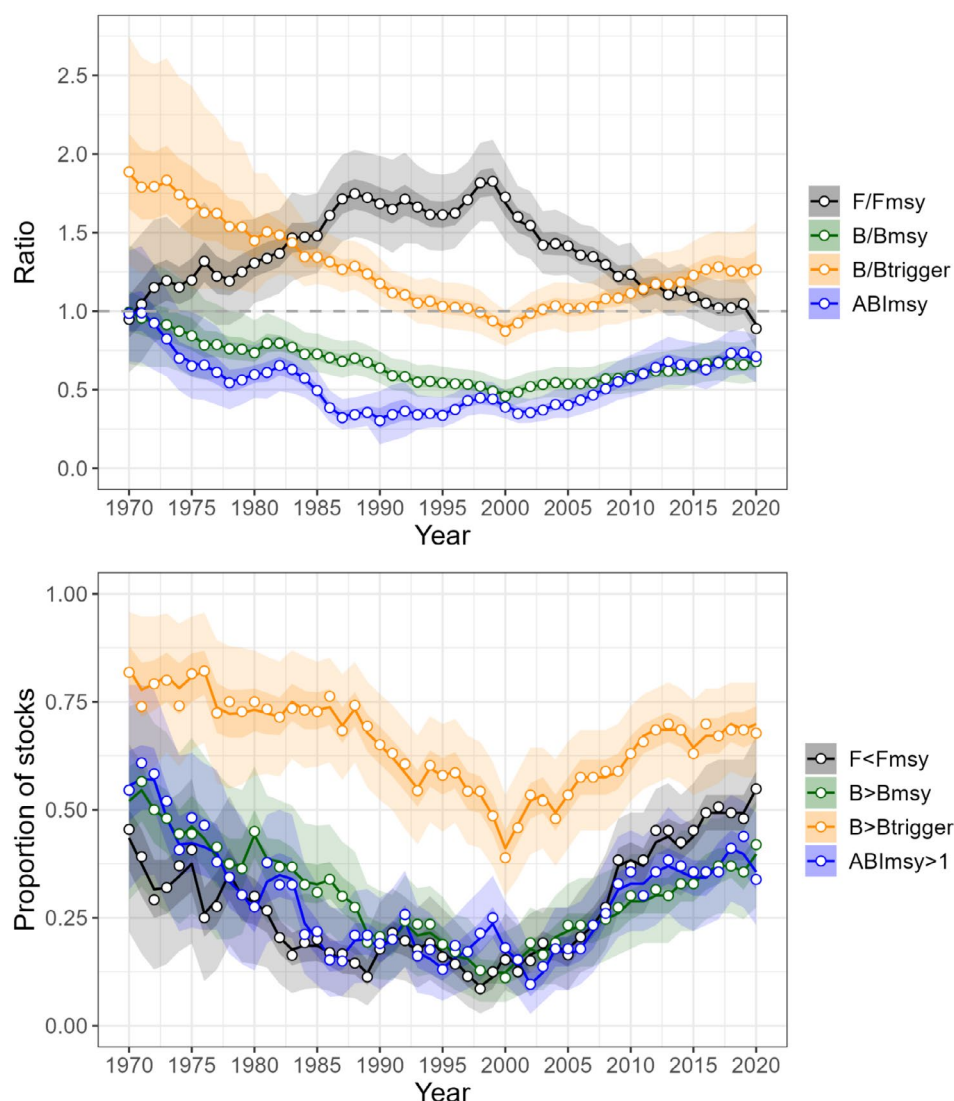


FIGURE 2 | Estimated median ratio of F/F_{MSY} , $B/B_{trigger}$, B/B_{MSY} and ABI_{MSY} across all 73 stocks (top) and the proportion of stocks for which F is below F_{MSY} , SSB is above B_{MSY} and $B_{trigger}$, and ABI_{MSY} is above 1 (bottom). In all cases, the line displays the estimated median and the estimated uncertainty is presented using both the 50% (darker shading; 25th and 75th percentiles) and the 95% (lighter shading; 2.5th and 97.5th percentiles) confidence intervals. The white points show the observed geometric mean across all stocks for a given indicator in a given year. The grey dashed horizontal line in the top panel signifies equivalence between the indicator of interest (e.g., SSB) and its equivalent reference point.

The perception of relative biomass levels differs, however, depending on whether biomass status is assessed by $B/B_{trigger}$ or B/B_{MSY} . When using $B_{trigger}$ as a proxy for B_{MSY} , stock status classification is a lot more optimistic, with the estimated median ratio only falling below 1 between 1998 and 2002, increasingly thereafter to a final value of 1.3 in 2020. In comparison, the estimated median ratio of B/B_{MSY} has never exceeded 1 during the 50-year time period, and although it indicates some level of rebuilding since 2000, biomass levels still remain at values below 70% of B_{MSY} (68% of B_{MSY} in 2020). Our results show that $B_{trigger}$ is on average (median) 61% of B_{MSY} , albeit the relationship between the two differs substantially across stocks (Figure S3). For instance, $B_{trigger}$ is less than 20% of B_{MSY} for haddock (*Melanogrammus aeglefinus*) in the waters around Rockall (had.27.6b), Atlantic cod (*Gadus morhua*) in the western English Channel and southern Celtic Seas (cod.27.7e-k), and Atlantic cod in the offshore waters west of

Greenland (cod.21.1). Moreover, $B_{trigger}$ is higher than the estimated B_{MSY} in four stocks. The median estimated ratio between the different estimates of MSY $B_{trigger}$ and B_{MSY} is 0.61, 0.51 and 0.60 for the B_{MSY} SR, B_{MSY} BH and B_{MSY} Est, respectively (Table S2).

Despite the joint trajectories of F/F_{MSY} and $B/B_{trigger}$ being below and above 1 in 2020, respectively, the proportion of stocks that have a biomass larger than $B_{trigger}$ is only around 70%. This indicates that around 30% of stocks in the Northeast Atlantic remain outside safe biological limits and still need to recover. In fact, during the last 5 years of the time period, half of the stocks were still being fished at levels that exceeded F_{MSY} . Moreover, the proportion of stocks that have a biomass that exceeds B_{MSY} and an age structure above ABI_{MSY} in 2020 is only 40% and 35%, respectively. It is also notable that the estimated trends in the proportion of stocks above or below their respective reference points

are highly comparable for $F < F_{MSY}$, $B > B_{MSY}$ and $ABI_{MSY} > 1$, whereas the estimated trend in proportion of stocks with $B > B_{trigger}$ is a lot higher throughout.

For all indicators, the uncertainty is estimated to be greater at the beginning of the 1970s and 1980s. This is because the number of observations differs as the length of the assessment time series differs between stocks and few stocks cover those earlier years. As shown in Figure S4, 95% data coverage in terms of having an estimate of stock status for all 73 stocks was only reached in the middle of the 1990s.

When aggregated by the three dominant taxonomic orders, the estimated median ratio of F/F_{MSY} shows the largest decline for pleuronectiformes, with the ratio being below 1 in the most recent years (F/F_{MSY} equal to 0.84 in 2020; Figure 3). Gadiformes also show a decreasing trend in the last two decades, although the estimated median ratio is still larger than 1.1 in 2020. For clupeiformes, the ratio has oscillated around 1 throughout the entire time series. The situation is rather different when analysing the trends in B/B_{MSY} . Although biomass has generally increased after 2000 for both pleuronectiformes and gadiformes, the median ratio is still under 1 for both taxonomic orders (0.51

for gadiformes and 0.87 for pleuronectiformes). In comparison, for clupeiformes B/B_{MSY} has declined since 2011. For all groups, $B/B_{trigger}$ shows a much more optimistic status than B/B_{MSY} , with $B/B_{trigger}$ in 2020 exceeding 1 for pleuronectiformes (1.47) and gadiformes (1.36) and being relatively close to 1 for clupeiformes (0.93).

ABI_{MSY} is found to track B_{MSY} with some evidence of a lag in both gadiformes and pleuronectiformes. Such a lag suggests that an increase in spawning biomass is not directly associated with an immediate increase in the proportion of older individuals as rebuilding of the age structure of a stock takes more time. For clupeiformes, ABI_{MSY} reached a value of 1 in 2009 and has remained stable in most recent years, notwithstanding a decline in spawning biomass. Given that the F/F_{MSY} has also remained stable, the decline in B/B_{MSY} may indicate that other processes than fishing mortality are important for maintaining stock biomass in this group.

The median ratio between F/F_{MSY} , $B/B_{trigger}$, B/B_{MSY} and ABI_{MSY} shows important differences between the major geographical regions of the Northeast Atlantic (Figure S5). Celtic, Arctic and North Sea stocks show a similar pattern in terms of F/F_{MSY} ,

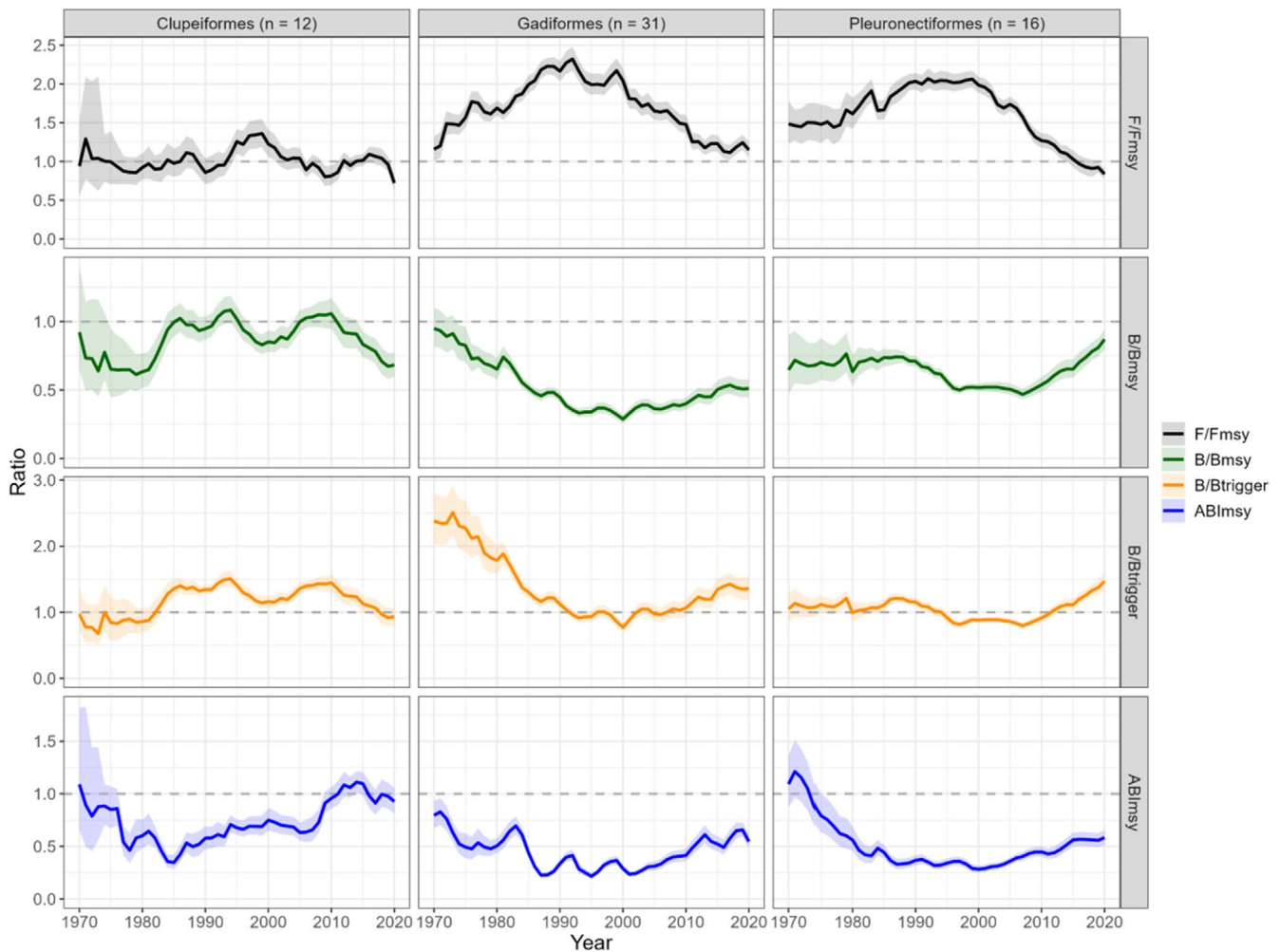


FIGURE 3 | Estimated median ratio of F/F_{MSY} , $B/B_{trigger}$, B/B_{MSY} and ABI_{MSY} by taxonomic order. In all cases, the estimated uncertainty is presented using 50% confidence intervals (shading; 25th and 75th percentiles). The grey dashed horizontal line in each panel signifies equivalence between the indicator of interest (e.g., SSB) and its equivalent reference point (e.g., B_{MSY}). Only the three most common taxonomic orders (Clupeiformes, Gadiformes and Pleuronectiformes) are shown as these make up most of the stocks in our database (81%).

which peaked in the 1990s and decreased thereafter, with F being below or very close to F_{MSY} in all three regions in 2020. $B/B_{trigger}$, B/B_{MSY} and ABI_{MSY} show corresponding trends, decreasing to a minimum level in the middle of the 1990s and increasing thereafter. However, it is notable that the increase is only substantial for Arctic stocks (with a $B/B_{trigger}$ value of 1.8 in 2020). Despite this, the median ratios do not reach B_{MSY} and/or ABI_{MSY} in 2020 in any of the three geographical regions. The situation in the Baltic Sea is rather different; F/F_{MSY} shows a relatively constant increase and B/B_{MSY} a general decline over the entire time series. On the other hand, ABI_{MSY} declines rapidly from large values in the 1970s and stays at low levels until 2020.

3.2 | Stock Status Classifications

The same joint median ratios and proportions are plotted using the FAO's SOFIA stock status classification and Kobe plots in Figure 4. Using $B_{trigger}$ as a surrogate for the B_{MSY} target reference point, the proportion of stocks classified according to FAO as either underfished or sustainably fished (Figure 4A, blue and green areas) is 77% in 2020. This drops to 53% when B_{MSY} is used and was as low as 23% in the late 1990s.

The proportion of stocks sustainably fished according to the Kobe plot (the green area in Figure 4B) also drops from around 45% to 30% when using B_{MSY} instead of $B_{trigger}$ in the most recent years. It is also notable that the number of stocks that are considered to be either rebuilding (yellow area) or overfished (red area) is much higher when comparing stock biomass to B_{MSY} as opposed to $B_{trigger}$. This is particularly true for stocks considered to be rebuilding in the most recent 10 years of our analysis, and our results indicate that there are currently around 30% of stocks that are above $B_{trigger}$ but have not yet rebuilt to levels expected under F_{MSY} .

When F/F_{MSY} is plotted against $B/B_{trigger}$, it shows that the estimated median of the two ratios has been in the overfishing section of the Kobe plot and not the overfished section for most of the last 50 years (Figure 4C). This would imply that although fishing levels have persistently exceeded F_{MSY} , biomass levels would have remained above the biomass target. In comparison, when F/F_{MSY} is plotted against B/B_{MSY} , the estimated medians show that the stocks have been overfished, and only in 2020 moved into a state of rebuilding in response to reduction of fishing pressure towards F_{MSY} levels.

3.3 | Projections

From the forward projection scenarios of the 73 stocks to the year 2030, we estimated that a continuation of current fishing levels will not rebuild the median ratio of B/B_{MSY} to a value above 1 (0.95 in 2030; Figure 5). However, a value of above 1 would be reached in the other three forecast scenarios. Specifically, the median ratio of B/B_{MSY} is predicted to reach 1.02 by 2030 if the ICES Advice Rule based on F_{MSY} and $B_{trigger}$ were to be implemented from 2023, whereas if $B_{trigger}$ is replaced by B_{MSY} , a value of 1.05 would be reached. Similarly, when using hindcasting to emulate a full implementation of the current ICES Advice Rule from 2014, the median ratio of B/B_{MSY} is predicted to have

exceeded 1 by 2023. Similar results are seen for ABI_{MSY} where values of 1 or close to 1 are reached under the three AR scenarios, but not the F_{sq} scenario (only reaches an ABI_{MSY} value of 0.9). However, for $B/B_{trigger}$, all four scenarios reach values by 2030 that exceed 1.5.

In terms of the proportion of stocks, almost all 73 stocks (97%) would have a biomass within safe biological limits (i.e., above $B_{trigger}$) by 2030 had the ICES Advice Rule been implemented and followed for all stocks since 2014 (Figure 6). This proportion is estimated to drop to around 95%, 95% and 90% of stocks for the AR 2023, AR B_{MSY} and F_{sq} scenarios, respectively. When using the FAO sustainability threshold, we find that 78% of stocks would have a biomass larger than 80% of B_{MSY} by 2030 under the F_{sq} scenario. This proportion rises to 99% for the AR 2014 and AR B_{MSY} scenarios, and around 96% for the AR 2023 scenario. In terms of age structure, fishing at F_{sq} will bring 68% of the stocks to an ABI_{MSY} target of 0.8 by 2030, increasing to 86% for the AR 2023 scenario, 95% for the AR B_{MSY} scenario and 99% for the AR 2014 scenario.

4 | Discussion

A systematic evaluation of the EU's Common Fisheries Policy indicated that it does not fully meet the criteria for an effective resource policy (Belschner et al. 2018) and that the original overarching target of the CFP of achieving F_{MSY} for all stocks by 2020 has not been met (this study). However, the most recent evaluation on the progress of the CFP describes a clear improvement in the status of Northeast Atlantic stocks during the last two decades, whereby the joint fishing mortality index for F/F_{MSY} for all stocks is closely approximating the F_{MSY} target (STECF 2023). In the Northeast Atlantic, most stocks have been through a process of rebuilding (Cardinale et al. 2012; Fernandes and Cook 2013; Froese et al. 2018; Zimmermann and Werner 2019). Similarly, a recent evaluation of the progress of fisheries management in the Northeast Atlantic depicts a positive situation with a general declining fishing mortality and increasing spawning biomass, with SSB being above $B_{trigger}$ on average (ICES 2019). Thus, the general status of Northeast Atlantic stocks as reported by the EU's STECF and ICES is currently estimated to be at or above target levels, both in terms of F but also in respect to levels of target biomass.

Despite the evident reduction in fishing mortality to average levels below F_{MSY} , our analysis shows that a large percentage of stocks are estimated to remain below B_{MSY} in 2020. We also found that the median spawning biomass in 2020 is around 68% of the biomass at B_{MSY} and that only around 50% of the stocks were classified to be above FAO's sustainable biomass threshold at 80% of B_{MSY} . These findings span 73 of the most commercially important stocks in the Northeast Atlantic, which in 2019, collectively contributed over 86% of all commercial catches in the region (Griffiths et al. 2023). The main difference between our findings and those reported by ICES and FAO's SOFIA is the use of a direct estimate of B_{MSY} as opposed to $B_{trigger}$, which based on our estimates is on average (median) 61% of our B_{MSY} estimates (Figure S3). Such differences call into question whether the use of $B_{trigger}$ as a biomass target (ICES 2019, 2023a) is consistent with the CFP, as $B_{trigger}$ is significantly below the biological

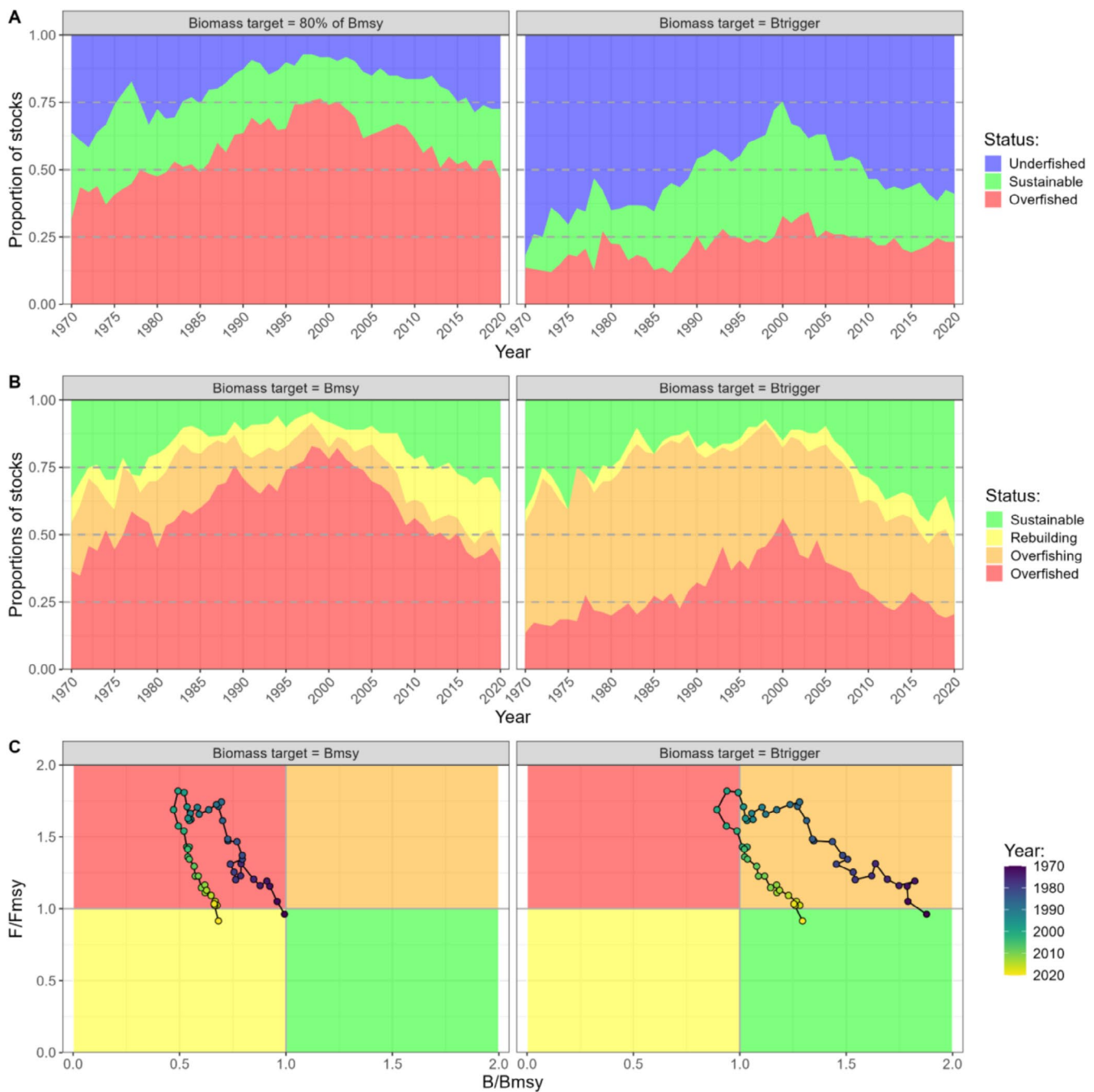


FIGURE 4 | FAO (A) and Kobe (B, C) stock status classification plots. In panels A and B, the proportion of stocks in each classification is presented, whereas in panel C, the estimated median B/B_{MSY} is plotted against the estimated median F/F_{MSY} . The biomass target (B/B_{MSY} , 80% of B/B_{MSY} or $B/B_{trigger}$) used to define a sustainable state is detailed in each panel. Dashed horizontal lines are provided at proportions of 0.25, 0.5 and 0.75 in panels A and B to aid interpretation.

B_{MSY} , and by its definition is the lower limit of biomass fluctuating around the target reference point B_{MSY} . Using the lower threshold as a surrogate of the target in this way leaves no buffer and it is therefore unlikely to ‘maintain populations of fish stocks above biomass levels capable of producing maximum sustainable yield’.

The discrepancies we observe in trends of B/B_{MSY} , $B/B_{trigger}$ and F/F_{MSY} have also been noted by others. For instance, Hilborn et al. (2021) described the ‘paradox’ that mean trends in the biomass of groundfish stocks in the Northeast Atlantic appeared to

increase above biomass targets in the early 2000s, despite fishing pressure being greater than F_{MSY} . The authors hypothesised that either F/F_{MSY} is systematically overestimated or there is a systematic bias in the use of $B_{trigger}$ as a proxy in the absence of available B_{MSY} estimates. Our results confirm a systematically over-optimistic stock classification, confirming that not only is the estimated median biomass ratio of $B/B_{trigger}$ a lot more optimistic than B/B_{MSY} , but also that the proportion of stocks considered to be in a state of rebuilding or in an overfished state differs substantially based on what biomass target is used. In fact, in 2020 we find that 20% of stocks are considered to be rebuilding

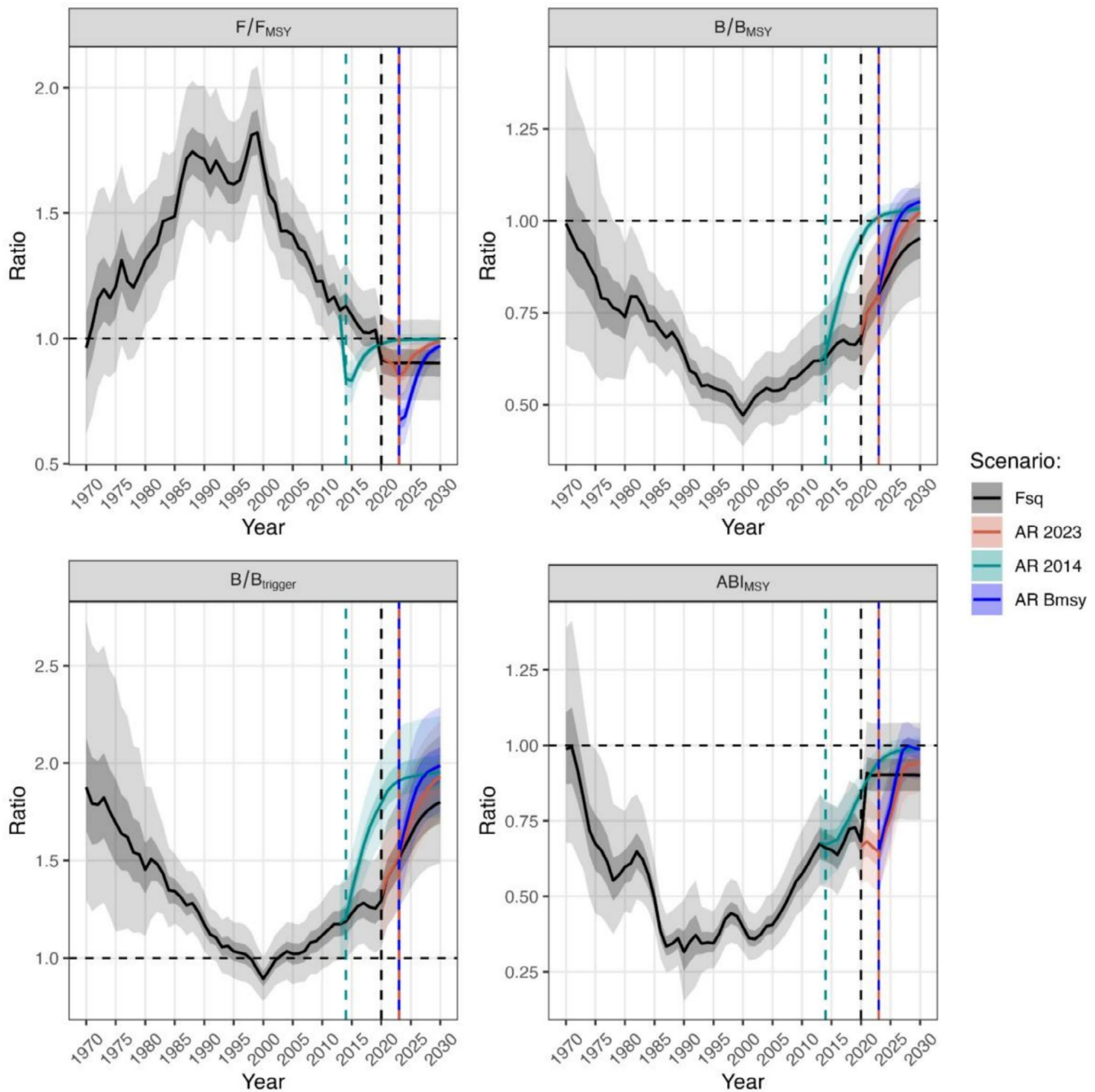


FIGURE 5 | Estimated median ratio of F/F_{MSY} , B/B_{MSY} , $B/B_{trigger}$ and ABI_{MSY} across all 73 stocks. Four scenarios were considered when forecasting to the year 2030: (1) F status quo (F_{sq} ; black), (2) use of the ICES Advice Rule (AR) assuming full implementation starting from 2014 (AR 2014; teal) or (3) 2023 (AR 2023; red), and (4) use of the AR but with B_{MSY} as an alternative biomass threshold for fishing at F_{MSY} since 2023 (AR B_{MSY} ; blue). As in other plots, the estimated uncertainty is presented using both the 50% (darker shading; 25th and 75th percentiles) and the 95% (lighter shading; 2.5th and 97.5th percentiles) confidence intervals. The black dashed horizontal line in each panel signifies equivalence between the indicator of interest (e.g., SSB) and its equivalent reference point (e.g., B_{MSY}). In comparison, the coloured dashed vertical lines (teal, black, red and blue) demonstrate when each forecast scenario was implemented.

when using B_{MSY} compared to 9% when using $B_{trigger}$ based on the Kobe plot classifications. Moreover, when using $B_{trigger}$ very few stocks (<5% in many years) have ever been classified as being in a state of recovery, but around 30% of stocks were found to have a biomass above $B_{trigger}$ but below B_{MSY} in 2020. This difference is of particular interest from a fisheries management perspective for two reasons. First, the conclusion that very few stocks have been in a state of recovery over the last 50 years is

counterintuitive considering that commercial fisheries started in the region during the 1800s (e.g., Cardinale et al. 2014; Engelhard et al. 2014) and many stocks have been exploited for more than a century prior to the start of our analysis. Further, even in the 1980s and 1990s, most stocks (>85%) were still being exploited at rates above F_{MSY} . Thus, it is likely that many stocks will have gone through a period of recovery from these past levels of exploitation and given the general declining trend in

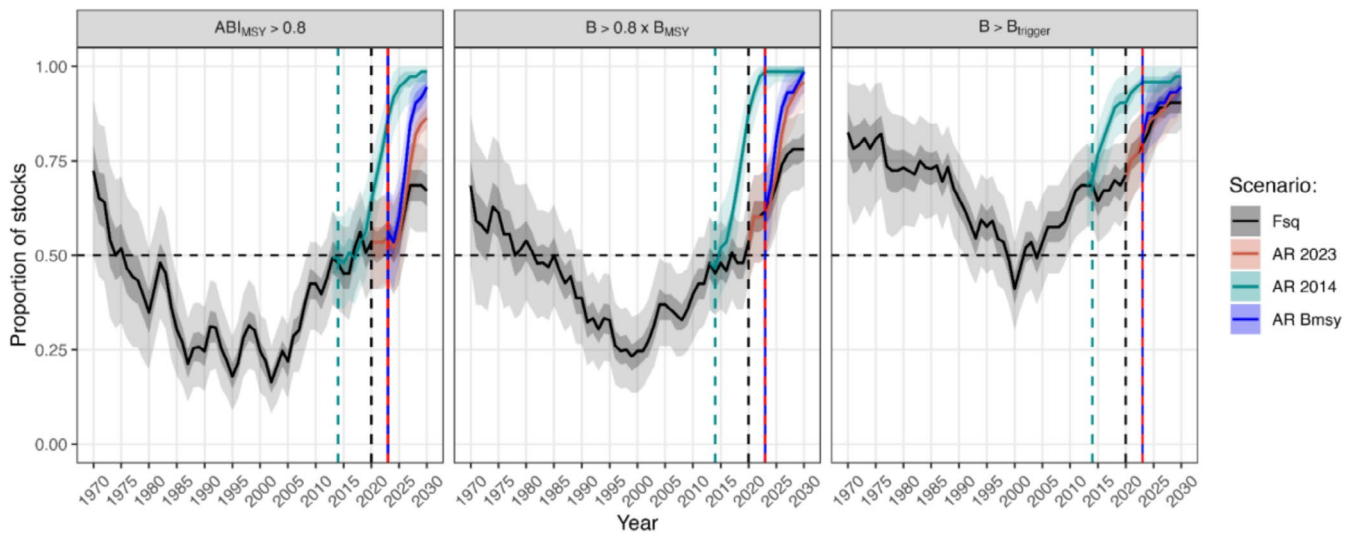


FIGURE 6 | Proportion of stocks that will have an ABI_{MSY} value above 0.8 (threshold used in Griffiths et al. 2023), an SSB above 80% of B_{MSY} (threshold used by the FAO) and an SSB above $B_{trigger}$ (threshold used by ICES). Four scenarios were considered when forecasting to the year 2030: (1) F status quo (F_{sq} ; black), (2) use of the ICES Advice Rule (AR) assuming full implementation starting from 2014 (AR 2014; teal) or (3) 2023 (AR 2023; red), and (4) use of the AR but with B_{MSY} as an alternative biomass threshold for fishing at F_{MSY} since 2023 (AR B_{msy} ; blue). As in other plots, the estimated uncertainty is presented using both the 50% (darker shading; 25th and 75th percentiles) and the 95% (lighter shading; 2.5th and 97.5th percentiles) confidence intervals. The black dashed horizontal line in each panel signifies a proportion of 0.5 and is provided to aid interpretation. In comparison, the coloured dashed vertical lines (teal, black, red and blue) demonstrate when each forecast scenario was implemented.

F/F_{MSY} , been experiencing a drop in fishing mortality and an increase in stock biomass (the definition of the rebuilding state in the Kobe plot) in recent years (Zimmermann and Werner 2019). Second, it also demonstrates how stocks might still be in a state of recovery from past exploitation, and although they are benefiting from $F/F_{MSY} < 1$, they are yet to build sufficient biomass to reach B_{MSY} or maintain fishing at F_{MSY} . This interpretation of stock status would be lost if using $B_{trigger}$ as opposed to B_{MSY} as the biomass target for stock classification.

Our findings on $B_{trigger}$ and B_{MSY} also highlight broader stock advice challenges. The ICES Advice Rule uses $B_{trigger}$ as a threshold for fishing at F_{MSY} , below which the management action of a linear reduction in F/F_{MSY} is triggered (ICES 2023a). In the absence of feedback control against a target, there is increased risk associated with bias in the assessment (e.g., Edgar et al. 2024), environmental variability (e.g., Rouyer et al. 2014) and complex interactions between them (Bjørnstad et al. 2004), meaning the biomass could fluctuate around the threshold or fail to achieve the B_{MSY} target. In most cases, biomass has yet to reach the B_{MSY} target. This shows that without an explicit feedback control rule based on biomass as well as F targets, it is not achievable to adequately assess the progress with respect to the MSY objective. Particularly, if B_{lim} is set too low as shown by WKREF1 for many stocks (ICES 2022), then B_{lim} as a safeguard will not be effective and this may lead to increased risk of SSB falling outside safe biological limits, which can be associated with rebuilding costs and forgoing future catch opportunities. Fisheries management requires the establishment of targets, limits and thresholds, whereas limit and target reference points serve for the purpose of stock status classification. However, thresholds such as $B_{trigger}$, trigger management action can be operationalised to optimise catches and reduce ecological risk and should, as such, not be considered for stock status classification.

This study also highlighted the advantage of considering alternative stock status indicators to provide additional inference about the age structure of commercially exploited stocks. In stock assessment and advice, we often assume that a given level of biomass will ensure that there are sufficient larger and older fish to maintain the expected level of productivity (Hixon et al. 2014; Barnett et al. 2017; Marshall et al. 2021). However, recent work (Griffiths et al. 2023) using a novel age-based indicator (ABI_{MSY}) has indicated that 62% of Category 1 stocks in the Northeast Atlantic are currently lacking older fish despite having biomasses at or above $B_{trigger}$ or F values below F_{MSY} . We find comparable trends here, with the estimated median ratio of ABI_{MSY} being around 0.68 in 2020, with values below 0.5 between 1985 and 2007. We also observe complementary trends between ABI_{MSY} and B/B_{MSY} , and inverse trends with F/F_{MSY} . To some extent, this is not surprising, because all three indicators are derived based on the underlying principle that a given biomass or age structure is needed to maintain a given level of catch. It is notable, however, that changes in ABI_{MSY} lagged to changes in B/B_{MSY} and F/F_{MSY} , as seen in both the JARA analysis by taxonomic order and the forward simulations to 2030. This is likely due to the time needed to re-establish or reduce a stock's age structure compared to its biomass, which can be interpreted as a buffer, especially if a stock is long living and has a longer generation time. Such lags call for a temporal component to be added to the evaluation of stock status when it comes to age or size and highlight a potential need for the CFPs aims on sustainable exploitation and stock rebuilding to be brought in line with the MSFDs Descriptor 3 for GES with includes an age and size component (EU-COM 2008, 2017). Despite this, it is important to note that changes in age structure may not always be expected to respond to changes in biomass or fishing mortality, as the likelihood of fish growing to older ages will depend

on other factors such as fisheries selectivity, resource availability and environmental conditions (e.g., Vasilakopoulos et al. 2020; Lindmark et al. 2022). Moreover, although age structure is not necessarily being controlled solely by changes in F , we do expect that a higher level of biomass will enable more fish to survive to older ages and larger sizes. Thus, B_{MSY} provides a greater buffer than B_{trigger} and is, in theory, better placed to facilitate stock recovery in terms of both biomass and age or size structure.

From a taxonomic perspective, none of the three major orders in our database (gadiformes, pleuronectiformes and clupeiformes) are currently above B_{MSY} , although clupeiformes did have a spawning biomass greater than B_{MSY} in the early 1990s and between 2006 and 2010. Gadiformes in comparison are estimated to have never had a median B/B_{MSY} above 1 between 1970 and 2020. This is despite recent declines in F/F_{MSY} and biomass values that have exceeded B_{trigger} and is maybe not surprising given the commercial importance of a species like Atlantic cod to the North Sea fishing industry over much of the last century (Engelhard et al. 2014). Pleuronectiformes have, however, shown a clear increase in spawning biomass in the most recent years, a trend that is highly correlated with a large decrease in F/F_{MSY} , albeit studies have also linked changes in the abundance and spatial distribution of flatfish species in the North Sea (e.g., European plaice *Pleuronectes platessa* and common sole *Solea solea*) to changes in both abiotic and biotic conditions, namely temperature, eutrophication and habitat suitability (Engelhard et al. 2011; Teal et al. 2012). Large scale differences in stock status were also evident by geographical regions, a trend that is also highlighted by Fernandes and Cook (2013) and Froese et al. (2018) for Northeast Atlantic stocks and by Hilborn et al. (2020) when analysing stocks in different US regions. In three of the four regions tested, we observe a clear decrease in F/F_{MSY} and an increase (Celtic Seas and Arctic waters) or a reasonably flat trend in B/B_{MSY} (Greater North Sea) since the turn of the century. The outlier to this trend occurs in the Baltic Sea, where F/F_{MSY} is estimated to have remained high and B/B_{MSY} has slowly declined. The estimated trend in ABR_{MSY} is also quite drastic in the Baltic Sea and mirrors the findings of Griffiths et al. (2023) and others (e.g., Mion et al. 2021; Mion et al. 2022), where stocks like eastern Baltic cod (cod.27.24-32) have experienced large truncations in their age and size structure, as well as reductions in biological rates and spatial range, due to a combination of pressures including historic fishing levels, temperature and oxygen content. It is also notable that none of the geographical regions included in our analysis are estimated to have a median biomass in 2020 above B_{MSY} . This is in line with the results by taxonomic order and suggests that the achievement of policy objectives like those detailed in CFP might require different levels of management action based on what type of species is being considered or which part of the Northeast Atlantic the species occurs in.

All studies have their limitations and these need to be acknowledged. For instance, we have assumed that the relationship between spawning biomass and recruitment (the SRR) can be described by a segmented regression or a Beverton-Holt in all 73 stocks. The segmented regression is the most used approach within ICES (e.g., ICES 2017, 2022), however, other options are available (e.g., Ricker), as are a mixture of SRRs. Consequently,

we recommend that similar comparisons are made on a stock-specific level in the future, especially when assumptions about SRRs are changed or evaluated. It is also notable that our analysis only runs to 2020, and it is probable that the assessment of some stocks may have changed in recent years, including updates to estimates of stock status and reference points. Reference points in ICES are periodically updated, either due to revisions in the assessment model or to better reflect current productivity (e.g., recruitment survivorship, mean weight, selectivity and others). Comparing historical exploitation rates and SSB to currently defined reference points has the potential to create biased indicators of stock status. Although this might be a potential source of bias in the historical trends (as reconstruction of reference point changes over time is seldom possible), there is no easy way around this issue. Currently, there is no official process to update the ICES FLR stock assessment database; however, ongoing developments around assessment transparency and reproducibility do provide possible solutions (e.g., the ICES Transparent Assessment Framework; <https://www.ices.dk/data/assessment-tools/Pages/transparent-assessment-framework.aspx>), and may enable regular updates of the database and our analysis.

Inconsistencies may also arise in our analysis because we have used ICES official benchmark estimates of F_{MSY} to derive the corresponding B_{MSY} , assuming a segmented regression SRR with a fixed breakpoint corresponding to B_{lim} . We chose to do this because ICES does not currently provide estimates of B_{MSY} , and in the absence of complete documentation of stock-specific benchmark configurations, we have used generic methodology to approximate B_{MSY} . To further evaluate the sensitivity of our B_{MSY} estimates, we have conducted additional sensitivity analyses using an alternative Beverton-Holt SRR to derive B_{MSY} (Table S2). This analysis involved the evaluation of both the official ICES F_{MSY} values and a direct estimate of F_{MSY} that maximises the surplus-production curve (F_{MSY} Est). In general, correlations among the alternative estimates of B_{MSY} across stocks were highly correlated and approximated a 1:1 linear relationship on the log scale (Figure S1). On average, the direct estimates of B_{MSY} (using F_{MSY} Est) were slightly larger than the B_{MSY} values derived from the reported ICES values. This difference likely reflects the fact that the direct estimates are not constrained by the precautionary fishing mortality $F_{\text{P0.5}}$ corresponding to a 5% probability that SSB falls below the limit reference point B_{lim} . Despite the strong overall correlation, the different B_{MSY} values also showed notable deviations around the 1:1 relationship, indicating stock-specific sensitivity to stock-recruitment assumptions. We therefore recommend that stock-specific B_{MSY} estimates are formally derived and fully documented within ICES benchmark working groups to improve internal consistency, enhance precision, and ensure reproducibility in the future.

Despite our results on stock status classification for stocks in the Northeast Atlantic being less optimistic than those recently conducted based on B/B_{trigger} by STECF and ICES, our projections for the year 2030 are promising. We estimated based on stock-specific forecasts that a perfect implementation of the ICES Advice Rule will allow the median ratio of B/B_{MSY} to exceed 1, independent of whether the implementation was started in 2014 or 2023. This

is also true if B_{trigger} is replaced by B_{MSY} in the ICES Advice Rule starting in 2023, with the only real difference being the increased speed at which the biomass target is reached. There are two mechanisms that can lead to B/B_{MSY} exceeding 1 when fishing at F_{MSY} . First, in Eqsim, B_{MSY} is derived as the median long-term SSB from stochastic projections at F_{MSY} , whereas our projections show the deterministic expectation of SSB, which is expected to be higher than the stochastic median but is consistent with the short-term deterministic forecasts used to set fishing opportunities within ICES. For our comparative analysis, we standardised stock-recruitment and forecast assumptions so that each F_{MSY} proxy (e.g., ICES PA-based or directly estimated F_{MSY}) has a corresponding deterministic B_{MSY} proxy, yielding $B/B_{\text{MSY}} = 1$ at equilibrium regardless of the scale of F_{MSY} . In contrast, Eqsim assumes lognormal recruitment with varying degrees of first-order autocorrelation (AR1), which typically yields a right-skewed distribution for SSB. Given the emergent properties of nonlinear population dynamics in stochastic age-structured simulations, the median SSB can be expected to be lower than the deterministic SSB trajectory based on mean recruitment, especially for stocks with high recruitment variance and autocorrelation. As a result, $B/B_{\text{MSY}} > 1$ can arise even when fishing exactly at F_{MSY} , and this divergence between the deterministic forecast of SSB and stochastic B_{MSY} can be interpreted as an inherent property of multiplicative lognormal variability in stochastic age-structured stock dynamics rather than any internal inconsistency in the modelling assumptions. Second, the trajectories represent means across all stocks, and some stocks start from very high biomass levels ($\text{SSB} > 1.5\text{--}3 \times B_{\text{MSY}}$ in 2020; Figure S6) and therefore may not approach B_{MSY} when ‘fished down’ at F_{MSY} .

Overall, we estimate that over 90% of the 73 stocks considered here will have a biomass above both B_{trigger} and 80% of B_{MSY} by 2030 in all three ICES Advice Rule scenarios. Moreover, we expect the age structure of stocks to also improve, with around 85%, 95% and 100% of the stocks having an ABJ_{MSY} value above 0.8 by 2030 in the AR 2023, AR B_{MSY} , and AR2014 scenarios, respectively. Such results suggest that the continuation of current management efforts may facilitate further improvements in the status of stocks in the Northeast Atlantic. Many stocks will also meet or exceed the aims of the CFP by 2030, having a biomass that is equivalent to or above B_{MSY} and capable producing MSY.

5 | Conclusions

Despite recent progress, where fishing levels have been reduced to or below F_{MSY} in the Northeast Atlantic (Fernandes and Cook 2013; this study), many commercially exploited stocks are yet to rebuild the spawning biomass necessary to produce MSY on average. This conclusion is informative to the monitoring of stock status conducted by STECF and ICES for data-rich assessments and differs from recent, more optimistic stock status evaluations for the region (e.g., Hilborn et al. 2021; FAO 2022; Sharma et al. 2025), which is predominantly due to the use of B_{trigger} as a surrogate for the true biomass target B_{MSY} . Specifically, we estimate that B_{trigger} is on average (median) only 61% of B_{MSY} , indicating that stocks with spawning biomasses above B_{trigger} may still be recovering from past levels of exploitation. In our opinion, a wide-scale shift to the reporting of B_{MSY} in the Northeast Atlantic will help solve this discrepancy, and as

shown in this study, will reveal that the rebuilding of European stocks towards the targets of the CFP and the FAO is slower than previously thought. The reporting of B_{MSY} estimates will also help reconnect management action to management goals, such that decisions on F can be made in the context of a biomass target and not a trigger. It will also aid future evaluations of global stock status, as fish stocks in the Northeast Atlantic will be able to be compared in a consistent and coherent way to other areas.

We also advocate that it is not possible to reliably monitor trends in stock status with respect to achieving the MSY objective without a feedback control that evaluates biomass against a defined target. Biomass targets may play a pivotal role in the Ecosystem Approach to Fisheries Management as stock resilience to anthropogenic effects as climate change and the role that species play in ecosystem dynamics are directly linked to their abundance and age structure (Mildenberger et al. 2022; Eurich et al. 2024; Griffiths et al. 2023). Reaching a biomass target of B_{MSY} will likely require strict adherence to the MSY approach to fisheries management by European management authorities in the next decade. Our simulations indicate that many stocks in the Northeast Atlantic have the potential to reach B_{MSY} by 2030. Such results provide a source of optimism for scientists, managers and fishers alike, as the rebuilding of stocks towards B_{MSY} will not only sustain larger catches and lower fishing costs associated with higher catch-per-unit-effort, but it can also help maintain and restore ecosystem functioning and increase the resilience of the marine ecosystems to future changes.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The original database of stock assessment inputs and outputs ($n = 78$) is freely available via ICES as part of the WKREF1 working group. The updated data and R code used in this study is freely available via a dedicated GitHub <https://github.com/Henning-Winker/ICES-Bmsy-FaF>. The updated data is also available as an R package (<https://github.com/laurieKell/icesdata>).

References

- Barnett, L. A. K., T. A. Branch, R. A. Ranasinghe, and T. E. Essington. 2017. “Old-Growth Fishes Become Scarce Under Fishing.” *Current Biology* 27, no. 18: 2843–2848.e2. <https://doi.org/10.1016/j.cub.2017.07.069>.
- Belschner, T., J. Ferretti, H. V. Strehlow, et al. 2018. “Evaluating Fisheries Systems: A Comprehensive Analytical Framework and Its Application to the EU’s Common Fisheries Policy.” *Fish and Fisheries* 20: 97–109. <https://doi.org/10.1111/faf.12325>.
- Bessell-Brown, P., A. E. Punta, G. N. Tucka, J. Day, N. Klaer, and A. Penney. 2022. “The Effects of Implementing a ‘Dynamic B0’ Harvest

- Control Rule in Australia's Southern and Eastern Scalefish and Shark Fishery." *Fisheries Research* 252: 106306. <https://doi.org/10.1016/j.fishr.2022.106306>.
- Bjørnstad, O. N., R. M. Nisbett, and J. M. Fromentin. 2004. "Trends and Cohort Resonant Effects in Age-Structured Populations." *Journal of Animal Ecology* 73, no. 6: 1157–1167.
- Buckland, S. T., K. P. Burnham, and N. H. Augustin. 1997. "Model Selection: An Integral Part of Inference." *Biometrics* 53, no. 2: 603–618.
- Buckland, S. T., K. B. Newman, L. Thomas, and N. B. Koesters. 2004. "State-Space Models for the Dynamics of Wild Animal Populations." *Ecological Modelling* 171: 157–175. <https://doi.org/10.1016/j.ecolmodel.2003.08.002>.
- Caddy, J. F., and R. Mahon. 2004. "Current Usage of Fisheries Indicators and Reference Points and Their Potential Application to Management of Fisheries for Marine Invertebrates." *Canadian Journal of Fisheries and Aquatic Sciences* 61, no. 8: 1307–1324.
- Cardinale, M. 2011. "Fishery Reform: Most Stocks Secure." *Nature* 476: 282. <https://doi.org/10.1038/476282a>.
- Cardinale, M., V. Bartolino, H. Svedäng, A. Sundelöf, R. T. Poulsen, and M. Casini. 2014. "A Centurial Development of the North Sea Fish Megafauna as Reflected by the Historical Swedish Longlining Fisheries." *Fish and Fisheries* 16: 522–533. <https://doi.org/10.1111/faf.12074>.
- Cardinale, M., H. Doerner, A. Abella, et al. 2012. "Rebuilding EU Fish Stocks and Fisheries, a Process Under Way?" *Marine Policy* 39: 43–52.
- Cardinale, M., G. C. Osio, and G. Scarcella. 2017. "Mediterranean Sea: A Failure of the European Fisheries Management System." *Frontiers in Marine Science* 4: 72. <https://doi.org/10.3389/fmars.2017.00072>.
- Cardinale, M., H. Winker, A. Orio, C. A. Griffiths, V. Bartolino, and L. Kell. 2024. "Occurrence, Strength, and Impact of Density-Dependent Processes in Key Productivity Parameters of Northeast Atlantic Stocks." *Ecology and Evolution* 14: e70375. <https://doi.org/10.1002/ece3.70375>.
- Colloca, F., M. Cardinale, M. Giannoulaki, et al. 2011. "Rebuilding Mediterranean Fisheries: Toward a New Paradigm for Ecological Sustainability in Single Species Population Models." *Fish and Fisheries* 14: 89–109. <https://doi.org/10.1111/j.1467-2979.2011.00453.x>.
- Cox, S. P., A. R. Kronlund, and A. J. Benson. 2013. "The Roles of Biological Reference Points and Operational Control Points in Management Procedures for the Sablefish (*Anoplopoma fimbria*) Fishery in British Columbia, Canada." *Environmental Conservation* 40, no. 4: 318–328.
- DFO. 2009. "A Fishery Decision-Making Framework Incorporating the Precautionary Approach." <https://www.dfo-mpo.gc.ca/reports-rapports/regs/sff-cpd/precaution-eng.htm>.
- Edgar, G. J., A. E. Bates, N. C. Krueck, S. C. Baker, R. D. Stuart-Smith, and C. J. Brown. 2024. "Stock Assessment Models Overstate Sustainability of the World's Fisheries." *Science* 385, no. 6711: 860–865.
- Engelhard, G. H., J. K. Pinnegar, L. T. Kell, and A. D. Rijnsdorp. 2011. "Nine Decades of North Sea Sole and Plaice Distribution." *ICES Journal of Marine Science* 68: 1090–1104.
- Engelhard, G. H., D. A. Righton, and J. K. Pinnegar. 2014. "Climate Change and Fishing: A Century of Shifting Distribution in North Sea Cod." *Global Change Biology* 20: 2473–2483.
- EU-COM. 2008. "Directive 2008/56/EC of the European Parliament and of the Council of 17 June 2008 Establishing a Framework for Community Action in the Field of Marine Environmental Policy (Marine Strategy Framework Directive)." *Official Journal of the European Union* L164: 19–40.
- EU-COM. 2017. "Commission Decision (EU) 2017/848 of 17 May 2017 Laying Down Criteria and Methodological Standards on Good Environmental Status of Marine Waters and Specifications and Standardised Methods for Monitoring and Assessment, and Repealing Decision 2010/477/EU." *Official Journal of the European Union* L125: 43–74.
- Eurich, J. G., W. R. Friedman, K. M. Kleisner, et al. 2024. "Diverse Pathways for Climate Resilience in Marine Fishery Systems." *Fish and Fisheries* 25, no. 1: 38–59.
- FAO. 1995. "Precautionary Approach to Fisheries." FAO Fisheries Technical Paper No. 350. FAO. 262 p.
- FAO. 2022. "The State of World Fisheries and Aquaculture 2022. Towards Blue Transformation." The State of World Fisheries and Aquaculture (SOFIA). pp. 266.
- FAO. 2023. *The State of Mediterranean and Black Sea Fisheries 2023*. General Fisheries Commission for the Mediterranean. <https://doi.org/10.4060/cc8888en>.
- Fernandes, P. G., and R. M. Cook. 2013. "Reversal of Fish Stock Decline in the Northeast Atlantic." *Biology Letters* 23: 1432–1437. <https://doi.org/10.1016/j.cub.2013.06.016>.
- Froese, R., G. Coro, K. Kleisner, and N. Demirel. 2014. "Revisiting Safe Biological Limits in Fisheries." *Fish and Fisheries* 17: 193–209. <https://doi.org/10.1111/faf.12102>.
- Froese, R., H. Winker, G. Coro, et al. 2018. "Status and Rebuilding of European Fisheries." *Marine Policy* 93: 159–170. <https://doi.org/10.1016/j.marpol.2018.04.018>.
- Gras, M., P. Vasilakopoulos, A. Pierucci, et al. 2023. "Protocol for the Monitoring of the Common Fisheries Policy." *Publications Office of the European Union, Luxembourg* 2023: JRC136235.
- Griffiths, C. A., H. Wennhage, H. Winker, V. Bartolino, A. Orio, and M. Cardinale. 2023. "Including Older Fish in Fisheries Management: New Age-Based Indicators and Reference Points for Commercial Fish Stocks." *Fish and Fisheries* 25: 18–37. <https://doi.org/10.1111/faf.12789>.
- Hilborn, R., R. O. Amoroso, C. M. Anderson, et al. 2020. "Effective Fisheries Management Instrumental in Improving Fish Stock Status." *Proceedings of the National Academy of Sciences of the United States of America* 117: 2218–2224. <https://doi.org/10.1073/pnas.1909726116>.
- Hilborn, R., D. J. Hively, N. B. Loke, et al. 2021. "Global Status of Groundfish Stocks." *Fish and Fisheries* 22: 911–928.
- Hixon, M. A., D. W. Johnson, and S. M. Sogard. 2014. "BOFFFFs: On the Importance of Conserving Old-Growth Age Structure in Fishery Populations." *ICES Journal of Marine Science* 71, no. 8: 2171–2185. <https://doi.org/10.1093/icesjms/fst200>.
- ICES. 2012. "Report of the ICES Advisory Committee 2012." ICES Advice. Book 9. 389 pp.
- ICES. 2015. "Report of the Joint ICES-MYFISH Workshop to Consider the Basis for F_{MSY} Ranges for All Stocks (WKMSYREF3), 17–21 November 2014, Charlottenlund, Denmark." ICES CM 2014/ACOM:64. 156 pp.
- ICES. 2017. "Report of the Workshop to Consider F_{MSY} Ranges for Stocks in ICES Categories 1 and 2 in Western Waters (WKMSYREF4), 13–16 October 2015, Brest, France." ICES CM 2015/ACOM:58. 187 pp.
- ICES. 2019. "Workshop on North Sea Stocks Management Strategy Evaluation (WKNMSME)." ICES Scientific Reports 1:12. 378 pp. <https://doi.org/10.17895/ices.pub.5090>.
- ICES. 2022. "Workshop on ICES Reference Points (WKREF1)." ICES Scientific Reports. 4:2. 70 pp. <https://doi.org/10.17895/ices.pub.9749>.
- ICES. 2023a. "Advice on Fishing Opportunities (2023)." General ICES Advice Guidelines. Report. <https://doi.org/10.17895/ices.advice.22240624.v2>.

- ICES. 2023b. "Benchmark Workshop on Baltic Pelagic Stocks (WKBALTPEL)." ICES Scientific Reports. 5:47. 350 pp. <https://doi.org/10.17895/ices.pub.23216492>.
- ICES. 2024a. "Workshop on the Calculation and Evaluation of New Reference Points for Category 1–2 Stocks (WKNEWREF)." ICES Scientific Reports 6:100. 241 pp. <https://doi.org/10.17895/ices.pub.27905664>.
- ICES. 2024b. "Advice on Identifying Operational Indicators and Defining Usable Threshold Values (Reference Points) for Criterion D3C3 Under MSFD Decision (EU) 2017/848." Report of the ICES Advisory Committee. ICES Advice 2024, sr.2024.01. <https://doi.org/10.17895/ices.advice.25265284>.
- Kell, L. T., P. Levontin, C. R. Davies, et al. 2016. "The Quantification and Presentation of Risk." In *Management Science in Fisheries: An Introduction to Simulation-Based Methods*, edited by C. Edwards and D. Dankel, 348–374. Routledge.
- Kell, L. T., I. Mosqueira, P. Grosjean, et al. 2007. "FLR: An Open-Source Framework for the Evaluation and Development of Management Strategies." *ICES Journal of Marine Science* 64: 640–646.
- Kell, L. T., R. Sharma, T. Kitakado, et al. 2021. "Validation of Stock Assessment Methods: Is It Me or My Model Talking?" *ICES Journal of Marine Science* 78, no. 6: 2244–2255.
- Lassen, H., C. Kelly, and M. Sissenwine. 2014. "ICES Advisory Framework 1977–2012: From Fmax to Precautionary Approach and Beyond." *ICES Journal of Marine Science* 71, no. 2: 166–172. <https://doi.org/10.1093/icesjms/fst146>.
- Lindmark, M., A. Audzijonyte, J. L. Blanchard, and A. Gårdmark. 2022. "Temperature Impacts on Fish Physiology and Resource Abundance Lead to Faster Growth but Smaller Fish Sizes and Yields Under Warming." *Global Change Biology* 28: 6239–6253. <https://doi.org/10.1111/gcb.16341>.
- Mace, P. 2001. "A New Role for MSY in Single-Species and Ecosystem Approaches to Fisheries Stock Assessment and Management." *Fish and Fisheries* 2: 2–32. <https://doi.org/10.1046/j.1467-2979.2001.00033.x>.
- Mace, P. M., and I. J. Doonan. 1988. "A Generalised Bioeconomic Simulation Model for Fish Population Dynamics." New Zealand Fishery Assessment Research Document 88/4. Fisheries Research Centre, MAFFish, POB, 297.
- Marshall, D. J., M. Bode, M. Mangel, R. Arlinghaus, and E. J. Dick. 2021. "Reproductive Hyperallometry and Managing the World's Fisheries." *Proceedings of the National Academy of Sciences of the United States of America* 118, no. 34: e2100695118. <https://doi.org/10.1073/pnas.2100695118>.
- Methot, R. D., and C. R. Wetzel. 2013. "Stock Synthesis: A Biological and Statistical Framework for Fish Stock Assessment and Fishery Management." *Fisheries Research* 142: 86–99. <https://doi.org/10.1016/j.fishres.2012.10.012>.
- Mildenberger, T. K., C. W. Berg, A. Kokkalis, et al. 2022. "Implementing the Precautionary Approach Into Fisheries Management: Biomass Reference Points and Uncertainty Buffers." *Fish and Fisheries* 23, no. 1: 73–92.
- Mion, M., C. A. Griffiths, V. Bartolino, et al. 2022. "New Perspectives on Eastern Baltic Cod Movement Patterns From Historical and Contemporary Tagging Data." *Marine Ecology Progress Series* 689: 109–126.
- Mion, M., S. Haase, J. Hemmer-Hansen, et al. 2021. "Multidecadal Changes in Fish Growth Rates Estimated From Tagging Data: A Case Study From the Eastern Baltic Cod (*Gadus morhua*, Gadidae)." *Fish and Fisheries* 22: 413–427. <https://doi.org/10.1111/faf.12527>.
- Moesgaard Albertsen, C., T. Perälä, M. Cardinale, H. Winker, and V. Trijoulet. 2024. "Rebuilding and Reference Points Under Compensatory and Depensatory Recruitment: A Meta-Analysis of Northeast Atlantic Fish Stocks." *Fish and Fisheries* 26: 171–192. <https://doi.org/10.1111/faf.12872>.
- New Zealand Ministry of Fisheries. 2008. *Harvest Strategy Standard for New Zealand Fisheries*. New Zealand Government. <https://fs.fish.govt.nz/Page.aspx?pk=113&dk=16543>.
- New Zealand Ministry of Fisheries. 2011. *Operational Guidelines for New Zealand's Harvest Strategy Standard*. New Zealand Government. https://fs.fish.govt.nz/Doc/22847/Operational_Guidelines_for_HSS_rev_1_Jun_2011.pdf.ashx.
- Nielsen, A., and C. W. Berg. 2014. "Estimation of Time-Varying Selectivity in Stock Assessments Using State-Space Models." *Fisheries Research* 158: 96–101. <https://doi.org/10.1016/j.fishres.2014.01.014>.
- Pacoureau, N., C. L. Rigby, P. M. Kyne, et al. 2021. "Half a Century of Global Decline in Oceanic Sharks and Rays." *Nature* 589: 567–571. <https://doi.org/10.1038/s41586-020-03173-9>.
- Probst, W. N., A. Kempf, M. Taylor, I. Martinez, and D. Miller. 2021. "Six Steps to Produce Stock Assessments for the Marine Strategy Framework Directive Compliant With Descriptor 3." *ICES Journal of Marine Science* 78, no. 4: 1229–1240. <https://doi.org/10.1093/icesjms/fsaa244>.
- Punt, A. E., A. D. Smith, D. C. Smith, G. N. Tuck, and N. L. Klaer. 2014. "Selecting Relative Abundance Proxies for B MSY and B MEY." *ICES Journal of Marine Science* 71, no. 3: 469–483. <https://doi.org/10.1093/icesjms/fst162>.
- Rouyer, T., J. M. Fromentin, M. Hidalgo, and N. C. Stenseth. 2014. "Combined Effects of Exploitation and Temperature on Fish Stocks in the Northeast Atlantic." *ICES Journal of Marine Science* 71, no. 7: 1554–1562.
- Scientific, Technical and Economic Committee for Fisheries (STECF). 2023. "Monitoring of the Performance of the Common Fisheries Policy (STECF-Adhoc-23-01)." Publications Office of the European Union, Luxembourg, JRC133325. <https://doi.org/10.2760/361698>.
- Sharma, R., M. Barange, V. Agostini, et al. 2025. *Review of the State of World Marine Fishery Resources–2025*. Food & Agriculture Org.
- Sharma, R., H. Winker, P. Levontin, et al. 2020. "Assessing the Potential of Catch-Only Models to Inform on the State of Global Fisheries and the UN's SDGs." *Sustainability* 2021, no. 13: 6101. <https://doi.org/10.3390/su13116101>.
- Sherley, R. B., R. J. M. Crawford, B. M. Dyer, et al. 2019. "The Status and Conservation of the Cape Gannet *Morus capensis*." *Ostrich* 90: 335–346.
- Sherley, R. B., H. Winker, C. L. Rigby, et al. 2019. "Estimating IUCN Red List Population Reduction: JARA—A Decision-Support Tool Applied to Pelagic Sharks." *Conservation Letters* 13: 1–10.
- Silvar-Viladomiu, P., L. Batts, C. Minto, D. Miller, and C. Lordan. 2022. "An Empirical Review of ICES Reference Points." *ICES Journal of Marine Science* 2022, no. 79: 2563–2578. <https://doi.org/10.1093/icesjms/fsac194>.
- Silvar-Viladomiu, P., C. Minto, G. Halouani, et al. 2021. "Moving Reference Point Goalposts and Implications for Fisheries Sustainability." *Fish and Fisheries* 22: 1345–1358. <https://doi.org/10.1111/faf.12591>.
- Simmonds, E. J., E. Portilla, D. Skagen, D. Beare, and D. G. Reid. 2010. "Investigating Agreement Between Different Data Sources Using Bayesian State-Space Models: An Application to Estimating NE Atlantic Mackerel Catch and Stock Abundance." *ICES Journal of Marine Science* 67: 1138–1153. <https://doi.org/10.1093/icesjms/fsq013>.
- Teal, L. R., R. van Hal, T. van Kooten, P. Ruardij, and A. D. Rijnsdorp. 2012. "Bio-Energetics Underpins the Spatial Response of North Sea Plaice (*Pleuronectes platessa* L.) and Sole (*Solea solea* L.) to Climate Change." *Global Change Biology* 18: 3291–3305. <https://doi.org/10.1111/j.1365-2486.2012.02795.x>.
- United Nations. 1995. "UN Agreement for the Implementation of the Provisions of the United Nations Convention on the Law of the Sea of

10 December 1982 Relating to the Conservation and Management of Straddling Fish Stocks and Highly Migratory Fish Stocks, 1995.” http://www.un.org/Depts/los/convention_agreements/texts/fish_stocks_agreement/CONF164_37.htm.

United Nations. 2002. “Plan of Implementation of the World Summit on Sustainable Development.” http://www.un.org/esa/sustdev/documents/WSSD_POI_PD/English/WSSD_PlanImpl.pdf.

Vasilakopoulos, P., E. Jardim, C. Konrad, et al. 2020. “Selectivity Metrics for Fisheries Management and Advice.” *Fish and Fisheries* 21: 621–638. <https://doi.org/10.1111/faf.12451>.

Winker, H., S. Kupschus, C. Konrad, M. Gras, and A. Mannini. 2022. “Suitability Study of the Bayesian State-Space Model ‘JARA’ for Stock Status Indicator Estimation.” STECF-Adhoc-22-02 Publications Office of the European Union, Luxembourg, 2023, JRC129283. <https://doi.org/10.2760/514979>.

Winker, H., N. Pacoureaux, and R. B. Sherley. 2020. “JARA: Just Another Red-List Assessment.” *bioRxiv*: 42. Cold Spring Harbor Laboratory. <https://www.biorxiv.org/content/early/2020/06/17/672899>.

Zimmermann, F., and K. M. Werner. 2019. “Improved Management Is the Main Driver Behind Recovery of Northeast Atlantic Fish Stocks.” *Frontiers in Ecology and the Environment* 17: 93–99. <https://doi.org/10.1002/fee.2002>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** faf70066-sup-0001-AppendixS1.docx.